

Release 1 Functional Specification

1. Document purpose and functional boundaries

1.1 Purpose

This document defines the observable behavior of Release 1 from the user's point of view.

It controls:

- which screens exist
- how users move between screens
- what information appears
- what actions users can perform
- what happens after each action
- which document statuses users see
- how documents move between statuses
- how extracted values are presented and corrected
- how warnings and failures are communicated
- how approval and rejection work
- how approved information appears on customer records
- how previous approved policy versions are preserved
- how retries and reprocessing behave
- how audit history is presented
- how usage limits affect processing
- how controlled demonstration mode behaves
- what observable outcomes define successful behavior

A UX designer should be able to design all core screens from this document. A software architect should be able to identify the major product responsibilities. A developer should be able to understand expected behavior. A QA engineer should be able to derive test cases.

1.2 What this document does not control

This document does not define:

- technical architecture
- database structures
- APIs
- background processing technology
- authentication providers
- AI, OCR, PDF, email, or hosting providers
- repository structure
- deployment design
- visual styling
- exact colors, fonts, spacing, or animation
- implementation tasks
- Codex prompts
- subscription billing

- Voice AI behavior
- multiple organizations or workspaces
- advanced user roles

These decisions belong in later planning documents.

1.3 Relationship to the approved product definition

The approved **Release 1 Product Definition and Scope** remains the source of truth for product strategy.

This specification does not change:

- the document-first release strategy
- the target insurance agency user
- the supported document type
- the supported extraction fields
- the human-review requirement
- the separation between AI candidate values and approved policy information
- the need for traceability
- the controlled demonstration requirement
- the limited budget
- the decision to delay Voice AI

Where several behaviors could satisfy the approved scope, this document recommends the smallest behavior that supports a polished and believable workflow.

1.4 Approved assumptions

The following assumptions are treated as approved:

1. Release 1 supports one agency context.
2. Staff users have the same functional abilities.
3. Release 1 does not include custom roles.
4. A controlled demo user has additional restrictions.
5. One uploaded file represents one personal auto policy.
6. Only English-language U.S. personal auto declaration page PDFs are supported.
7. AI-generated information remains untrusted until human approval.
8. Approved policy versions are preserved rather than silently rewritten.
9. Customer contact information and document-extracted insured information may differ.
10. Customer acknowledgement email is optional and reviewer-triggered.
11. The product must remain demonstrable when live AI processing is unavailable.

2. Product navigation and information structure

2.1 Recommended primary navigation

Release 1 should contain five primary navigation items:

1. Overview

2. **Customers**
3. **Documents**
4. **Review Queue**
5. **Usage**

A separate Settings section should not be included.

2.2 Navigation decision

Recommendation: Include Overview, Customers, Documents, Review Queue, and Usage. Place sign-out, current user information, and demo-mode information in the account area rather than a Settings page.

Problem solved: Users need direct access to daily operational work without navigating through unrelated SaaS administration pages.

Simpler alternative: Use only Customers and Review Queue.

Why the simpler alternative is insufficient: Users would lack a central place to find failed documents, understand recent activity, or check whether processing limits have been reached.

Accepted trade-off: Five navigation items create slightly more interface surface, but each supports a distinct Release 1 responsibility.

2.3 Overview

Purpose

The Overview screen gives users an operational starting point.

Who uses it

Normal staff users and controlled demo users.

Information contained

- documents needing review
- documents currently processing
- documents that failed
- recently approved documents
- recent customer and document activity
- current processing-limit status
- direct actions to create a customer or upload a document

Why it belongs

It prevents users from searching through several pages to understand what needs attention.

2.4 Customers

Purpose

The Customers area is used to create, find, and inspect customer records.

Information contained

- manually entered contact information
- associated documents
- current approved policy
- previous approved policy versions
- customer-related audit history

Why it belongs

The customer record connects document processing to an actual business workflow. Without it, the product would behave like a disconnected extraction tool.

2.5 Documents

Purpose

The Documents area provides a complete operational list of uploaded documents, including completed, failed, rejected, and processing documents.

Information contained

- document name
- related customer
- upload date
- current status
- warning summary
- processing method
- reviewer and approval details where applicable

Why it belongs

The Review Queue contains only work awaiting human review. Users still need a place to locate failed, rejected, processing, and historical documents.

2.6 Review Queue

Purpose

The Review Queue contains documents that are ready for human review.

Information contained

- customer
- document
- warning severity
- time waiting
- latest processing result
- whether another user has completed or opened the review

Why it belongs

This screen represents the main daily operational task.

2.7 Usage

Purpose

The Usage screen shows processing activity and whether cost or volume limits are close to being reached.

Information contained

- documents processed
- pages processed
- OCR usage
- AI extraction attempts
- failed attempts
- replayed attempts
- estimated cost where sufficiently reliable
- configured limits
- current limit status

Why it belongs

Cost control is a Release 1 product requirement rather than an administrative extra.

2.8 Settings decision

Recommendation: Do not include a Settings navigation item in Release 1.

Problem solved: Avoids implying the product has configurable organization, provider, billing, role, or workflow settings that are not part of Release 1.

Simpler alternative: Add an empty Settings screen for future use.

Why the simpler alternative is insufficient: An empty or weak settings area makes the product feel unfinished.

Accepted trade-off: Any Release 1 operational limits or demo information are displayed in context rather than being configurable through a general settings interface.

2.9 Default landing page

After sign-in, the user lands on **Overview**.

The only exception is when the user was redirected to sign-in after attempting to access a protected page. After successful sign-in, the user returns to that original page when it is still valid.

2.10 Page titles

Primary page titles should be:

- Overview
- Customers
- Create Customer
- Edit Customer
- Customer Details
- Documents
- Upload Declaration Page
- Review Queue
- Review Document
- Usage

Customer and document pages should include identifying context in the visible heading, such as the customer's name or document file name.

2.11 Breadcrumb behavior

Breadcrumbs should be used on detail and workflow pages where the user may arrive from several locations.

Examples of conceptual paths include:

- Customers / Customer Name
- Customer Name / Documents / Document Name
- Review Queue / Customer Name / Document Name
- Documents / Document Name / Review

Breadcrumbs should not attempt to preserve every previous filter or navigation step. Returning through the browser or a clear "Back to Review Queue" action should preserve the previous list filters when practical.

2.12 Linking related information

- A customer name links to the customer detail screen.
 - A document name links to the document detail or review screen, depending on status.
 - A current policy links to its source document and approval information.
 - A historical policy version links to the document and reviewer that created it.
 - Audit entries link to the related document or customer when the target still exists.
 - A Needs Review document provides a direct Review action.
 - An approved document opens in a read-only review summary unless the user starts a correction or reprocessing workflow.
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3. Authentication and session behavior

3.1 Sign-in behavior

The sign-in screen contains:

- an identifier field
- a password or equivalent credential field
- a sign-in action
- a controlled demo entry action when demo access is enabled
- a short explanation that the product contains synthetic insurance data in demo mode

Successful sign-in sends the user to Overview or back to the originally requested protected page.

3.2 Invalid credentials

When credentials are invalid:

- access is denied
- entered secret information is cleared where appropriate
- the user sees a clear sign-in error
- the message does not reveal whether a specific account exists
- the user remains on the sign-in screen
- no protected information becomes visible

3.3 Protected pages

A user who attempts to open a protected page without a valid session is redirected to sign-in.

After successful sign-in:

- the user returns to the requested page when it still exists
- otherwise, the user lands on Overview

3.4 Sign-out behavior

Sign-out:

- ends the current authenticated session
- returns the user to sign-in
- prevents protected pages from being reopened through normal browser navigation
- does not delete uploaded documents or saved work

If unsaved reviewer changes exist, the user receives a warning before sign-out.

3.5 Expired sessions

When a session expires while the user is viewing a page:

- the current page remains visible only as long as necessary to explain that the session expired
- protected actions become unavailable

- the user is asked to sign in again
- after sign-in, the user returns to the same customer, document, or review page when possible

3.6 Session expiration during upload

If the session expires before upload submission:

- the upload does not begin
- the selected file may need to be selected again
- the user is told that no upload was created

If the session expires after the upload was accepted:

- background processing may continue
- after signing in again, the user can locate the document through Documents or the related customer
- the product must not create multiple uploads because the user repeats the action

3.7 Session expiration during review

If the session expires while the reviewer has unsaved edits:

- the product warns the user before redirecting where practical
- unsaved changes should be preserved in the current browser session where practical
- after sign-in, the user returns to the same review
- the system checks whether another user changed or completed the review
- unsaved changes are never applied without the user submitting them again

3.8 Session expiration during approval or rejection

A critical action is considered complete only when the user receives a confirmed result.

If the session expires before completion:

- approval or rejection must not be assumed
- after sign-in, the user sees the actual current document status
- if the action already completed, the user sees the completed result
- if it did not complete, the user may submit again
- repeated submission must not create duplicate approvals or audit events

3.9 Controlled demo access

Recommendation: Provide a clearly labelled demo entry that starts a restricted synthetic-data session.

Problem solved: Portfolio viewers need to explore the product without receiving normal staff capabilities or access to real data.

Simpler alternative: Share normal staff credentials.

Why the simpler alternative is insufficient: Shared full-access credentials could allow destructive actions, uncontrolled paid usage, and conflicting demo changes.

Accepted trade-off: Demo behavior requires a small number of restrictions and visible explanations.

3.10 Unsaved change behavior

Unsaved-change protection applies to:

- customer create and edit forms
- field edits in the review workspace
- approval notes or warning acknowledgements
- rejection notes
- email recipient changes

When the user navigates away with unsaved changes:

- the product warns that changes will be lost
 - the user may remain on the page or discard the changes
 - browser closing or refresh should receive equivalent protection where practical
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4. Overview screen

4.1 Overview decision

Recommendation: Include a small operational Overview screen.

Problem solved: Staff need to know what requires attention immediately after sign-in.

Simpler alternative: Open directly on the Review Queue.

Why the simpler alternative is insufficient: It hides failed processing, active processing, recent approvals, and usage-limit problems.

Accepted trade-off: The Overview adds one screen, but it contains no advanced reporting or decorative analytics.

4.2 Information shown

The Overview contains the following operational summaries.

Needs Review

Shows the number of documents currently in Needs Review.

Clicking opens the Review Queue with the default filters.

Processing

Shows the number of documents currently processing.

Clicking opens Documents filtered to Processing.

Failed

Shows the number of unresolved failed documents.

Clicking opens Documents filtered to Failed.

Approved Recently

Shows the number approved during the current recent period and a short list of the most recent approvals.

Clicking the count opens Documents filtered to Approved. Clicking an individual item opens its approved review summary.

Usage Status

Shows:

- current usage level relative to the configured period limit
- whether live processing is available
- whether the environment is using replayed results, restricted live processing, or normal live processing

Clicking opens Usage.

4.3 Recent activity

Recent activity should show a short chronological list of important events, such as:

- customer created
- document uploaded
- processing failed
- document became ready for review
- reviewer approved or rejected a document
- policy version became active
- acknowledgement email succeeded or failed

It should not include low-level processing details.

Clicking an item opens the related customer or document.

4.4 Primary calls to action

The Overview contains two primary actions:

- Create Customer
- Upload Document

Upload Document requires selecting or confirming a customer.

4.5 Empty state

When no customers or documents exist:

- the operational counts show zero
- recent activity explains that no activity exists yet
- the main action is Create Customer
- a secondary sample-demo action may be available in demo mode

4.6 Loading state

While data is loading:

- the page structure remains stable
- summary areas display loading placeholders
- actions that do not depend on loaded data remain available
- no zero counts appear until loading completes

4.7 Error state

If the Overview cannot load:

- a page-level error explains that the operational summary is unavailable
- a retry action is available
- primary navigation remains available
- the error does not imply that documents or customers were deleted

4.8 Partial data

If one summary cannot load but others can:

- available summaries remain visible
- the unavailable section shows a local error and retry action
- the entire page does not fail

5. Customer list

5.1 Displayed information

Each customer row shows:

- customer full name
- email when available
- phone number when available
- current approved policy carrier when available
- current policy number when available
- current policy expiration date when available
- number of associated documents
- an attention indicator when a document needs review or has failed
- last activity time

The list should not show sales pipeline, lead source, owner, revenue, or marketing fields.

5.2 Search behavior

Search should match:

- customer full name
- email
- phone number
- current policy number
- primary named insured from the current approved policy

Search results update after the user submits or pauses typing briefly.

Search is case-insensitive and ignores harmless formatting differences in phone numbers.

5.3 Filters

Release 1 includes only these customer filters:

- All customers
- Has documents needing review
- Has failed documents
- Has active approved policy
- No active approved policy

5.4 Sorting

Default sorting is most recent activity first.

Available sorting options:

- most recent activity
- customer name
- policy expiration date

Customers without a policy expiration date appear after dated records when sorting by expiration.

5.5 Pagination

Recommendation: Use page-based pagination for the customer list.

Problem solved: The user needs predictable navigation and a stable list when records grow.

Simpler alternative: Load every customer at once.

Why the simpler alternative is insufficient: It becomes slow and difficult to navigate as the portfolio grows.

Accepted trade-off: Pagination adds controls but is easier to understand than an endless list.

The current page, search, filters, and sorting should remain in the page address so users can return to the same view.

5.6 Create customer action

Create Customer is visible at the top of the list.

After successful creation, the user is taken to the new customer's detail screen.

5.7 Row click behavior

Clicking a row opens Customer Details.

Actions inside a row should not accidentally trigger row navigation.

5.8 Empty state

When no customers exist:

- explain that customers connect documents and approved policies
- offer Create Customer
- in demo mode, offer Load Sample Customer when permitted

5.9 No-results state

When search or filters return no results:

- show the active search and filters
- offer Clear Search and Filters
- do not show the general no-customer onboarding message

5.10 Loading state

Display stable loading rows without temporarily showing an empty list.

5.11 Failure state

If the customer list cannot load:

- show a page-level error
- allow retry
- keep Create Customer unavailable only if customer creation also depends on the failed service
- preserve the user's search and filter selections

5.12 Duplicate customer warning

A possible duplicate warning appears during customer creation, not as a hidden background action.

Potential duplicate matching may consider:

- similar full name

- matching email
- matching phone number
- matching current policy number

The warning should:

- display possible matching customers
 - allow the user to open an existing customer
 - allow the user to continue creating a new customer after explicit confirmation
 - never merge customers automatically
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6. Create and edit customer

6.1 Minimum manually entered fields

Recommendation:

- Full name: required
- Email: optional
- Phone number: optional
- Contact address: optional

At least one of email or phone should be encouraged but not required.

6.2 Decision explanation

Problem solved: A customer must exist before document processing, but staff may initially have only a name and a declaration page.

Simpler alternative: Require full name, email, phone, and address.

Why the simpler alternative is insufficient: It could block intake when the agency has not yet collected complete contact information.

Accepted trade-off: Some customer records will be incomplete, but the document workflow can begin without inventing contact data.

6.3 Staff-entered information

Staff-entered customer information represents contact information known to the agency.

It is editable independently from approved policy information.

6.4 Extracted information

Extracted named-insured and mailing-address information remains attached to the document review until approval.

It must not silently replace staff-entered customer information.

6.5 Approved policy information

After approval, the policy version shows the insured name and mailing address found in the document.

These values may differ from the contact information in the customer summary.

6.6 Required and optional validation

Full name

Required.

The form rejects:

- empty values
- values containing no meaningful name characters
- values that exceed the supported field length

The system should not enforce a fixed first-name and last-name structure.

Email

Optional.

When entered, it must be in a valid email format.

Invalid email prevents saving.

Phone number

Optional.

When entered, it must contain a plausible phone number.

The product may normalize display formatting but must not invent missing digits.

Contact address

Optional.

Release 1 should use a simple multiline address entry rather than a complex address-validation workflow.

6.7 Save behavior

On save:

- inline validation runs
- duplicate candidates are displayed when found
- the user may choose an existing customer or confirm creation

- a successful save opens Customer Details
- a temporary success message confirms creation or update
- an audit event records the customer creation or important change

6.8 Cancellation

Cancelling creation returns to the previous page or Customer List.

If entered data exists, the user receives a discard warning.

Cancelling edit returns to Customer Details without applying changes.

6.9 Partially created customers

Release 1 allows customers with only a full name.

The product should visibly indicate missing contact methods where relevant, especially before email acknowledgement.

It should not use a separate “draft customer” status.

6.10 Edit behavior

Staff may edit:

- customer full name
- email
- phone
- contact address

Editing customer contact information does not alter previous approved policy versions.

6.11 Save failure

If saving fails:

- entered information remains visible
- the user sees a clear error
- the user may retry
- a partially saved customer must not be presented as successfully created
- repeated save attempts must not create duplicate customers

6.12 Different insured name

When the document contains a named insured different from the customer’s staff-entered name:

- a critical customer-identity warning appears
- approval is blocked until the reviewer confirms that the document belongs to the selected customer
- the reviewer may correct the extracted insured name when extraction was wrong
- when the difference is legitimate, the reviewer must explicitly acknowledge it

- the staff-entered customer name is not automatically changed

6.13 Different mailing address

When the extracted mailing address differs from the customer's contact address:

- an important warning appears
 - the reviewer may confirm that the addresses are intentionally different
 - the reviewer may choose during approval to update the customer contact address
 - updating the contact address requires an explicit choice
 - the approved policy always retains the mailing address shown in the document
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7. Customer detail screen

7.1 Recommended structure

The Customer Detail screen should use five sections or tabs:

1. Summary
2. Current Policy
3. Documents
4. Policy History
5. Activity

7.2 Structure decision

Recommendation: Use tabs or clearly separated sections rather than one very long page.

Problem solved: Customer contact information, active policy information, documents, historical versions, and audit events are related but serve different tasks.

Simpler alternative: Place everything on one page.

Why the simpler alternative is insufficient: The page becomes difficult to scan once several documents and policy versions exist.

Accepted trade-off: Tabs add one navigation level but keep each responsibility clear.

7.3 Summary

Shows:

- staff-entered full name
- email
- phone
- contact address
- incomplete-contact indicators
- current approved policy summary
- documents needing attention

- recent customer activity

Available actions:

- Edit Customer
- Upload Document
- Open Current Policy
- Open Review for a document needing review

7.4 Current Policy

When an active approved policy exists, show:

- carrier
- policy number
- effective date
- expiration date
- total premium when present
- named insureds
- policy mailing address
- vehicles
- drivers
- supported coverage values
- source document
- reviewer
- approval date
- whether this version replaced a previous active version

Available actions:

- Open Source Review
- View Policy History
- Start Correction Review
- Send Acknowledgement when eligible and not already sent

Approved values are read-only.

7.5 No approved policy

When no approved policy exists:

- explain that no document has been approved
- show any documents in processing or review
- provide Upload Document
- avoid displaying blank policy fields as though they are missing customer data

7.6 Documents

Shows all documents associated with the customer, including:

- document name
- upload date

- status
- warning summary
- processing method
- reviewer
- approval or rejection date

Available actions depend on status:

- open processing details
- open review
- open approved summary
- retry failed processing
- inspect rejection
- upload replacement

7.7 Policy History

Shows previous approved versions in reverse approval order.

Each version displays:

- active, previous, or historical designation
- carrier
- policy number
- effective and expiration dates
- approval date
- reviewer
- source document
- replacement reason or relationship when available

7.8 Current and historical distinction

The current active policy is visually and textually marked as **Current**.

Older approved versions are marked **Previous** or **Historical**.

The product must not rely on color alone.

7.9 New approval replacing an active policy

When approval may replace an existing active policy:

- the approval screen compares the policy identity and dates
- the reviewer sees which version would become current
- replacement requires confirmation
- the previous active version remains in Policy History
- the old version is never deleted or rewritten

7.10 Activity

Shows customer-understandable audit events related to:

- customer information
- documents
- processing
- review
- policy approvals
- rejections
- corrections
- email acknowledgements

Technical diagnostic details remain outside this activity view.

8. Document upload workflow

8.1 Upload entry points

Upload can start from:

- Overview
- Customer Details
- Documents

When upload starts from Customer Details, that customer is already selected.

When upload starts elsewhere, the user must search for and select a customer before selecting or submitting the file.

8.2 One-file decision

Recommendation: Release 1 accepts one file per upload workflow.

Problem solved: One-file processing makes customer association, duplicate handling, errors, progress, and paid-processing confirmation easier to understand.

Simpler alternative: Allow batch upload.

Why the simpler alternative is not actually simpler: Batch upload creates partial failures, several customer associations, mixed statuses, and less understandable cost confirmation.

Accepted trade-off: Staff must repeat the upload action for each policy, but the Release 1 workflow remains reliable and clear.

8.3 File selection

The upload screen supports:

- a file-selection action
- drag and drop on desktop
- replacement of the selected file before submission
- clear display of file name and file size

Selecting several files at once results in a message that Release 1 accepts one file at a time. The user may choose one of the selected files.

8.4 Supported file type

Only PDF files are accepted.

A non-PDF file:

- is rejected before upload
- does not consume processing usage
- produces a clear supported-format message

File extension alone must not be treated as proof that the file is a valid PDF.

8.5 File-size limit

Recommendation: Display and enforce a Release 1 maximum of 15 MB per file.

Problem solved: Prevents unusually large files from increasing upload, storage, OCR, and AI costs.

Simpler alternative: Accept any PDF size.

Why the simpler alternative is insufficient: It creates uncontrolled cost and unreliable processing.

Accepted trade-off: Some unusually large declarations must be compressed or replaced before upload.

When a file exceeds the limit:

- submission is blocked
- no document record enters processing
- the user is told to provide a smaller PDF

8.6 Page limit

Recommendation: Support up to 10 pages per uploaded PDF.

Problem solved: Personal auto declarations normally fit within a limited page count, while larger files may contain several unrelated policy documents.

Simpler alternative: No page limit.

Why the simpler alternative is insufficient: It weakens the one-policy-per-file rule and increases paid processing risk.

Accepted trade-off: A user may need to separate an oversized file before uploading.

If the PDF exceeds the page limit:

- processing does not start
- the user sees the detected page count
- the user is asked to upload only the relevant declaration pages

8.7 Password-protected or encrypted PDFs

When the PDF requires a password or cannot be opened for reading:

- the upload is rejected before paid extraction where possible
- the user is told to upload an unlocked copy
- the password is not requested or stored in Release 1
- no AI processing begins

8.8 Corrupted PDFs

When the file cannot be parsed or rendered:

- the document enters Failed only if upload had already completed
- otherwise, the selection is rejected before submission
- the user sees that the file appears damaged
- the recovery action is to upload another copy

8.9 No readable pages

When the PDF opens but contains no processable pages:

- processing stops
- the document becomes Failed
- the reason states that no readable pages were found
- no candidate extraction result is created
- the user may upload a replacement

8.10 One-policy-per-file expectation

Before submission, the upload screen explains that the file should contain one personal auto policy.

If processing detects several likely policies or conflicting policy numbers:

- the document may continue to Needs Review
- a critical multiple-policy warning appears
- approval is blocked until the reviewer confirms one policy or rejects the document
- the system must not silently combine policies

8.11 Customer association

The selected customer is displayed before submission.

Changing the customer after file selection is allowed before upload begins.

After upload begins:

- customer association cannot be changed through the upload screen
- a wrong association is handled through rejection and a new upload under the correct customer
- Release 1 does not provide a general document-transfer feature

8.12 Upload progress

During file transfer, show:

- file name
- percentage or meaningful progress indication
- Cancel Upload while cancellation remains possible
- explanation that processing begins after transfer

Upload progress is distinct from document processing progress.

8.13 Cancel upload

If the user cancels before the file is accepted:

- the transfer stops
- no processing begins
- no lasting document appears in Documents
- no paid usage occurs

If the file has already been accepted:

- cancellation may no longer be available
- the product explains that processing has begun
- the user can later reject the result if the upload was unnecessary

8.14 Upload interruption

When connection is lost before upload completes:

- show an interrupted-upload message
- allow the user to retry the upload
- avoid creating several documents from repeated attempts
- preserve the selected customer where practical

8.15 Duplicate detection

After the file is accepted but before paid processing begins, the system checks for an exact or strongly matching previous upload.

A possible duplicate warning shows:

- related customer
- previous document
- upload date
- previous status
- whether the previous document was approved
- whether another processing attempt would use paid services

Available actions:

- Open Existing Document
- Cancel New Upload
- Continue as Intentional New Upload

Continuing requires explicit confirmation.

8.16 Duplicate across another customer

If the matching file belongs to another customer:

- the warning is critical
- the user sees that the same file already exists elsewhere
- normal processing is paused
- the user may cancel and inspect the other customer
- continuing requires explicit confirmation and an explanation that the association should be checked carefully

8.17 Successful upload

After a successful upload:

- the document enters Processing
- the user sees a document summary with the selected customer
- processing progress is visible
- the user may leave the page
- the document appears in Documents and on Customer Details
- an audit event records the upload

8.18 Redirect after upload

Recommendation: Redirect to the document processing screen rather than directly back to the customer.

Problem solved: The user receives immediate confirmation and understands that processing has started.

Simpler alternative: Return directly to Customer Details.

Why the simpler alternative is insufficient: The user may not understand whether processing began successfully.

Accepted trade-off: The user performs one extra navigation step if they want to return to the customer.

9. Document lifecycle and status model

9.1 Recommended user-visible statuses

Release 1 should use five main document statuses:

1. Processing
2. Needs Review
3. Approved
4. Rejected
5. Failed

Uploading is a temporary screen state rather than a lasting document status.

Queued, Extracting Text, Running OCR, Extracting Fields, and Validating should appear as processing descriptions under Processing rather than separate top-level statuses.

9.2 Status decision

Recommendation: Use one Processing status with a detailed progress description.

Problem solved: Users need to understand progress without learning internal processing stages.

Simpler alternative: Display only “Processing” with no detail.

Why the simpler alternative is insufficient: Long-running documents would appear stuck.

Alternative: Expose every internal stage as a separate status.

Why that alternative is excessive: It creates noisy lifecycle transitions and ties the interface to implementation details.

Accepted trade-off: The product shows helpful stage descriptions while preserving a small stable status model.

9.3 Processing

Meaning

The document has been accepted and is undergoing validation, text extraction, OCR when needed, AI extraction, or result validation.

Entry

A successful new upload or approved retry enters Processing.

User-visible progress descriptions

Possible descriptions include:

- Waiting to process
- Checking file
- Reading document text
- Reading scanned pages
- Extracting policy information
- Checking extracted information
- Taking longer than expected
- Reprocessing with previous approved policy preserved

These descriptions are informational, not separate statuses.

Available actions

- View customer
- Leave page
- View processing details
- Cancel only when the initial file transfer has not completed
- No review or approval action

Exit

Processing moves to:

- Needs Review when a usable candidate result exists
- Failed when processing cannot produce a reviewable result
- Approved or Rejected only after later human action

Finality

Not final.

9.4 Needs Review

Meaning

Processing produced a reviewable candidate result.

Entry

The document enters Needs Review after extraction and validation complete.

Available actions

- Open Review
- View customer
- Inspect warning summary
- Reject
- Start review edits

- Approve only after review requirements are satisfied

Unavailable actions

- Directly update customer policy without review
- Send acknowledgement email
- Start another processing attempt while an active review result exists, unless the reviewer deliberately chooses reprocessing

Exit

- Approved
- Rejected
- Processing when intentional reprocessing starts

Finality

Not final.

9.5 Approved

Meaning

A staff user completed review and approved a policy version.

Entry

Only through the approval workflow.

Available actions

- View approved result
- View source document
- View audit history
- Send acknowledgement email
- Start correction review
- Start intentional reprocessing
- Upload a newer or replacement document

Unavailable actions

- Direct editing of approved values
- Repeated approval of the same result
- Silent replacement of the active policy

Exit

The document itself generally remains Approved.

If a correction review or reprocessing starts:

- the existing approval remains visible

- the screen shows a secondary activity state
- the active customer policy remains unchanged until another approval occurs

Finality

The approved result is final and preserved, but a new approved version may later replace it.

9.6 Rejected

Meaning

A reviewer decided the document should not create or update an approved policy version.

Entry

Only through the rejection workflow.

Available actions

- View rejection reason
- View source document
- Upload replacement
- Start intentional reprocessing when the reason permits
- View audit history

Unavailable actions

- Approve the rejected result directly
- Send acknowledgement email
- Update customer policy from the rejected result

Exit

Rejected remains a preserved final outcome. A new processing attempt may create a new review result without erasing the rejection.

Finality

Final for that review result.

9.7 Failed

Meaning

The system could not produce a usable review result.

Entry

Examples include:

- corrupted PDF
- unreadable pages

- OCR failure
- AI provider timeout after allowed handling
- invalid extraction structure
- unsupported document detected before a reviewable result exists
- processing exceeded the allowed completion window

Available actions

Depending on failure type:

- Retry Technical Processing
- Upload Replacement
- View Failure Summary
- View Customer
- View Previous Attempts

Unavailable actions

- Approve
- Send acknowledgement email
- Update customer policy

Exit

- Processing through an allowed retry
- remains Failed when retries are unavailable or fail
- a replacement upload creates a separate document

Finality

Not necessarily final, but the failed attempt is preserved.

9.8 Unsupported document

Unsupported document should be treated as a failure category, not a separate top-level status.

The document shows:

- Status: Failed
- Reason: Unsupported document type

This avoids expanding the status model while keeping the reason clear.

9.9 Cancelled upload

A transfer cancelled before acceptance does not create a lasting document status.

If cancellation occurs after acceptance is no longer possible, processing continues and the user may reject the result later.

9.10 Delayed or stuck processing

A document remains Processing while it is within the allowed processing window.

When it exceeds the normal expected duration:

- display “Taking longer than expected”
- reassure the user that leaving the page will not stop processing
- do not automatically encourage repeated uploads
- show the most recent confirmed progress time

When the maximum processing window is exceeded:

- the document becomes Failed
- the reason identifies a processing timeout or unconfirmed completion
- a controlled technical retry may become available

9.11 Provider timeout

A provider timeout may be retried automatically only within the product’s controlled processing rules.

From the user’s point of view:

- brief internal recovery remains Processing
- repeated or final timeout becomes Failed
- the failure summary explains that the external processing service did not complete
- the user can use a limited technical retry

9.12 Reprocessing

Reprocessing is a named action, not a separate permanent document status.

While reprocessing:

- the document shows Processing or a secondary “Reprocessing” indicator
- previous attempts remain available
- an existing approved policy remains active
- previous reviewer corrections are preserved
- the new result must be reviewed independently

9.13 Approved document processed again

An approved document being reprocessed must not appear as though its approval was undone.

The screen shows:

- Existing approved policy remains active
- Reprocessing in progress

When the new result completes:

- the document indicates that a new review result is available

- the current approved policy remains active
- another approval is required before any customer policy changes occur

If reprocessing fails:

- the approved result remains valid
 - the failed attempt is shown in history
 - the document continues to display its approved outcome with a failed-reprocessing notice
-

10. Documents list

10.1 Purpose

Documents is the complete operational record of uploaded files.

It includes:

- processing documents
- documents awaiting review
- approved documents
- rejected documents
- failed documents

10.2 Displayed information

Each row shows:

- document file name
- related customer
- upload date and time
- main status
- processing method when known
- warning count and highest warning severity
- reviewer when reviewed
- approval or rejection date when applicable
- latest processing attempt result
- duplicate indicator when relevant

10.3 Search

Search matches:

- document file name
- customer name
- policy number from the latest reviewable or approved result
- carrier name

10.4 Filters

Available filters:

- status
- highest warning severity
- processing method
- upload date range
- reviewed or not reviewed
- duplicate warning
- live processing or replayed result in demo mode

10.5 Sorting

Default sorting is newest upload first.

Available sorting:

- newest upload
- oldest upload
- most recent activity
- customer name
- status
- approval date

10.6 Actions

Row actions depend on status:

- Processing: View Progress
- Needs Review: Review
- Approved: View Approved Result
- Rejected: View Rejection
- Failed: View Failure or Retry
- All: Open Customer

10.7 Row click

Clicking the row opens the document's most relevant detail:

- processing detail for Processing
- review workspace for Needs Review
- approved summary for Approved
- rejection summary for Rejected
- failure summary for Failed

10.8 Empty state

When no documents exist:

- explain the supported document type
- provide Upload Document

- link to create a customer if no customers exist

10.9 No-results state

Show:

- active search and filters
- Clear Filters
- no suggestion that documents were deleted

10.10 Loading state

Show stable placeholder rows and preserve existing filter controls.

10.11 Failed-list state

When filtered to Failed:

- each row shows a short failure category
- retryable failures are distinguished from replacement-required failures
- the screen explains that failed documents did not update customer policies

10.12 Difference from Review Queue

Documents is a complete history and operational search screen.

Review Queue is an action-focused screen containing only documents currently ready for human review.

Failed documents do not appear in the Review Queue because they do not yet contain a reviewable result.

11. Review queue

11.1 Purpose

The Review Queue shows documents that require human review before they can affect customer policy information.

11.2 Which documents appear

Only documents with a current reviewable result in Needs Review appear.

Failed, approved, rejected, and processing documents do not appear.

An approved document with a newly completed reprocessing result appears because a new review decision is required. It must be marked as having an existing approved policy still active.

11.3 Ordering priority

Default order:

1. documents with critical warnings
2. documents with important warnings
3. documents with no blocking warnings
4. within each group, oldest waiting first

This prioritizes risk and aging without introducing formal service-level rules.

11.4 Displayed information

Each item shows:

- customer name
- document file name
- upload time
- time waiting for review
- carrier and policy number when available
- highest warning severity
- warning count
- OCR-used indicator
- existing approved-policy indicator
- latest reviewer activity when relevant

11.5 Filters

Available filters:

- warning severity
- OCR used
- existing approved policy
- age range
- customer

11.6 Search

Search matches:

- customer name
- document file name
- carrier
- policy number

11.7 Sorting

Available sorting:

- priority and oldest first
- newest first
- oldest first

- customer name
- warning count

11.8 Open-review action

Clicking Review opens the Human Review Workspace.

The previous queue search, filters, and sort should be preserved for return navigation.

11.9 Empty state

When no documents require review:

- clearly state that the queue is clear
- show links to Documents and Upload Document
- do not show failed documents as though they were completed work

11.10 Stale items

When a queue item has waited beyond a reasonable operational period:

- show an age indicator
- do not create a formal breach or escalation state
- continue ordering older items earlier within their warning priority

11.11 Another user already reviewed the document

If another user approves or rejects the document:

- it disappears from the queue
- a user who opens an outdated queue item is taken to the completed result
- the product explains that another user completed the review

11.12 Another user currently viewing

Release 1 does not require assignment or hard locking.

When recent activity indicates another user may be reviewing:

- show a non-blocking indicator
 - allow the second user to open the document
 - prevent stale approval or stale edits from overwriting completed work
 - show current status before any final action
-

12. Human review workspace

12.1 Purpose

The review workspace must help the reviewer answer five questions:

1. What did the system extract?
2. Where did each value come from?
3. What appears missing, invalid, or conflicting?
4. What did the reviewer change?
5. Is the result safe to approve?

12.2 Desktop layout

Recommendation: Use a two-panel desktop layout.

- Left panel: source PDF
- Right panel: extracted fields, warnings, and review actions

The action area remains visible or easily reachable without covering the document.

Problem solved: Reviewers must compare candidate values against the source without switching pages.

Simpler alternative: Show the PDF and form on separate screens.

Why the simpler alternative is insufficient: It increases memory burden and slows correction.

Accepted trade-off: The review workspace requires a larger desktop viewport and more careful responsive behavior.

12.3 Narrow-screen layout

On narrower screens:

- Source and Extracted Data become two clear views or tabs.
- The user can switch without losing edits.
- Selecting evidence opens the relevant source page.
- The approval summary remains accessible.
- The product does not attempt to display both panels at unreadably small widths.

12.4 Mobile behavior

Full review and approval are not required on small mobile screens.

Mobile users may:

- inspect status
- read warning summaries
- view customer details
- open a simplified document preview

Editing and approval may display a message recommending desktop or tablet use.

12.5 Review header

The header shows:

- customer
- document name
- upload time
- current status
- processing method
- latest attempt
- existing active policy where applicable
- warning summary
- whether the page may be stale
- link back to Review Queue

12.6 Source document viewer

Page navigation

The viewer supports:

- current page and total page count
- next page
- previous page
- direct page selection
- field-to-page navigation

Zoom

The viewer supports:

- zoom in
- zoom out
- fit page
- fit width

Loading

While pages load:

- the field panel remains usable
- a local viewer loading state appears
- approval is not blocked solely because one page preview is still loading if evidence can otherwise be reviewed

Failure to render

If the PDF cannot render:

- show a persistent viewer error

- allow download when permitted
- continue displaying extracted fields and evidence text
- approval receives at least an important warning
- if the source cannot be inspected by any available method, approval is blocked

Download availability

Normal staff users may download the original source document.

Demo users may download only approved synthetic sample files when demo policy allows.

Field-to-page navigation

Selecting a field's page evidence:

- moves the viewer to the supporting page
- highlights the evidence area only when reliable location information exists
- otherwise shows the supporting excerpt without pretending an exact location is known

Bounding boxes are optional and not required for Release 1.

Evidence excerpts

Evidence excerpts should:

- show enough surrounding text to understand the value
- identify the source page
- avoid presenting the entire extracted document text by default
- distinguish direct PDF text from OCR-derived text when useful

Rotated pages

If the page was detected or displayed as rotated:

- the user can rotate the view
- the processing summary indicates that orientation handling occurred or may be needed
- a low-quality orientation issue may create an OCR concern warning

Scanned pages

Scanned or OCR-processed documents show a visible "OCR used" indicator.

12.7 Field groups

Fields are organized in this order:

1. Policy
2. Named Insured
3. Premium
4. Vehicles
5. Drivers
6. Coverage

Each group shows:

- completion state
- warning count
- whether reviewer edits exist
- expansion and collapse behavior when useful

12.8 Policy group

Contains:

- insurance carrier name
- policy number
- effective date
- expiration date

12.9 Named insured group

Contains:

- primary named insured
- mailing address
- additional named insureds

12.10 Premium group

Contains:

- total policy premium

The group explains that premium is optional when a clear total is not present.

12.11 Vehicles group

Each vehicle appears as a separate item containing:

- year
- make
- model
- VIN
- comprehensive deductible when present
- collision deductible when present

12.12 Drivers group

Each driver appears as a separate item containing:

- full name

12.13 Coverage group

Contains:

- bodily injury liability limit
- property damage liability limit
- uninsured or underinsured motorist limit when present

Vehicle-specific deductibles remain inside each vehicle rather than being duplicated here.

12.14 Field presentation

A field may display:

- current candidate or reviewer-edited value
- original AI candidate value when the reviewer changed it
- source representation when formatting was normalized
- supporting text excerpt
- supporting page
- validation state
- warning severity
- OCR concern
- not-present state
- optional reviewer note
- edit history summary

12.15 Raw value behavior

The interface should not display raw provider output.

When the source representation differs meaningfully from the normalized field value, the review view may show both.

Example behavior:

- source representation: "\$1,250.00"
- normalized candidate: "1,250.00 USD"

The product should not expose model formatting noise that does not help review.

12.16 Entering edit mode

A field enters edit mode when the reviewer selects Edit or focuses the editable control.

Edit mode shows:

- current editable value
- expected format guidance when useful
- Save
- Cancel

12.17 Saving an edit

On Save:

- format validation runs
- invalid values remain in edit mode with an inline explanation
- valid values replace the displayed candidate
- the field is marked Reviewer Corrected
- the original candidate remains visible in field history
- related warnings are recalculated or marked resolved
- the edit is included in the approval summary

12.18 Cancelling an edit

Cancel restores the last saved review value.

It does not restore the original AI value when a previous reviewer edit had already been saved.

12.19 Clearing a value

Clearing a value is different from marking it not present.

When the reviewer clears a value:

- the field becomes unresolved
- any required-field warning appears
- the reviewer must either enter a value or mark it not present where allowed

12.20 Marking not present

Mark Not Present means the reviewer checked the document and believes the value is not included.

The action:

- requires confirmation for important or critical fields
- records the reviewer decision
- removes “unreviewed missing” status
- does not invent a value
- may still block approval for fields that are required to identify the policy

12.21 Undoing a recent correction

A reviewer may undo the most recent unsaved or saved correction before approval.

Undo restores the previous review value, not necessarily the original AI candidate when several edits occurred.

After approval, the version is immutable and cannot be undone directly.

12.22 Multiple edits

Multiple edits to the same field are preserved in audit history at an understandable level.

The final approval summary displays:

- original candidate
- final reviewer value

It does not need to display every intermediate keystroke.

12.23 Adding a missed vehicle

The reviewer may add a vehicle when it appears in the document but was not extracted.

The reviewer enters the supported vehicle fields and identifies supporting page evidence where practical.

The added vehicle is marked Reviewer Added.

12.24 Removing an incorrect vehicle

The reviewer may remove an extracted vehicle when it does not belong to the policy.

The original extracted item remains in audit history.

Removal requires a short confirmation because it changes a list item rather than a single field.

12.25 Adding a missed driver

The reviewer may add a driver with a full name and supporting page reference where practical.

The item is marked Reviewer Added.

12.26 Removing an incorrect driver

The reviewer may remove an incorrectly extracted driver.

The removal is preserved in audit history.

12.27 Reordering vehicles or drivers

Recommendation: Do not support manual reordering in Release 1.

Problem solved: None of the approved business behavior depends on display order.

Simpler alternative: Preserve document order automatically.

Why it is sufficient: Reviewers can identify each item through its values.

Accepted trade-off: Users cannot arrange vehicles or drivers for preference, but the workflow avoids unnecessary list-management behavior.

12.28 Unsaved changes

Unsaved changes are visibly indicated.

If the user attempts to:

- leave the page
- open another customer
- return to the queue
- sign out
- refresh

the product asks whether to discard changes or stay.

12.29 Warning levels

Release 1 uses three warning levels.

Informational

Meaning:

- the reviewer should be aware of a condition
- no action is required

Examples:

- OCR was used
- optional field was not present
- source formatting was normalized

Approval:

- not blocked

Important

Meaning:

- the value may be incomplete, weakly supported, or different from the customer record
- the reviewer must inspect it

Examples:

- mailing address differs
- optional premium has several candidates
- partial VIN appears to be intentionally masked
- evidence is weak

Approval:

- blocked until the reviewer corrects, marks not present, or explicitly acknowledges the warning

Critical

Meaning:

- policy identity, customer association, or internal consistency is not trustworthy

Examples:

- conflicting policy numbers
- expiration before effective date
- named insured may belong to another customer
- several policies may be in one file
- unsupported document
- no inspectable source

Approval:

- blocked until corrected
- acknowledgement is allowed only for specifically permitted identity differences where the reviewer confirms the document belongs to the selected customer
- some critical warnings require rejection rather than acknowledgement

12.30 Warning acknowledgement

Acknowledgement requires:

- a deliberate action
- a short reason for critical permitted acknowledgements
- no note for simple important warnings unless context is needed

Acknowledgement does not change the source value.

12.31 Readiness summary

The action area displays:

- number of unresolved critical warnings
- number of unresolved important warnings
- acknowledged warnings
- reviewer-corrected fields
- reviewer-added or removed items
- missing optional values
- whether approval is currently available

12.32 Approval readiness

The document is Ready for Approval when:

- no unresolvable critical warning remains
 - all important warnings are corrected, marked not present, or acknowledged
 - required policy identity fields are complete
 - the reviewer has confirmed customer association
 - there are no unsaved edits
 - the current processing result has not changed
 - the document is still in Needs Review
-

13. Validation and uncertainty behavior

13.1 General rule

Validation helps the reviewer focus attention. It does not declare the document unquestionably correct.

Evidence strength, format validity, cross-field consistency, OCR quality, and reviewer action should be shown separately where practical.

13.2 Carrier name

Expected behavior

Display a readable carrier name supported by document text.

Common invalid conditions

- missing
- several different carrier names
- agency name mistaken for carrier
- unreadable OCR result
- value supported only by weak text

Missing behavior

Missing carrier is critical and blocks approval until entered or the document is rejected.

Conflict behavior

Several plausible carriers create a critical warning.

Evidence expectation

At least one supporting page and text excerpt.

Acknowledgement

Conflict cannot be acknowledged without choosing or entering the correct carrier.

13.3 Policy number

Expected behavior

Display one policy number associated with the declaration page.

Common invalid conditions

- missing
- several policy numbers
- OCR substitution
- value appears to be an account, quote, or form number
- policy number differs across pages

Missing behavior

Critical. Approval blocked.

Conflict behavior

Critical. Reviewer must select or enter the correct number.

Evidence expectation

Supporting page and nearby policy-label text.

Acknowledgement

Not sufficient. A policy number must be resolved.

13.4 Effective and expiration dates

Expected behavior

Both dates should be present and form a valid period.

Common invalid conditions

- missing date
- invalid calendar date
- ambiguous ordering
- expiration before effective date
- several policy periods
- date extracted from a billing or print date

Missing behavior

Critical for either date.

Conflict behavior

Critical when several periods appear.

Evidence expectation

Both dates should have supporting text.

Acknowledgement

Dates must be corrected. A reversed or impossible period cannot be approved through acknowledgement.

13.5 Named insured**Expected behavior**

At least one primary named insured should be identified.

Common invalid conditions

- missing
- agent or driver mistaken for named insured
- several possible primary insureds
- name conflicts with selected customer
- OCR-damaged name

Missing behavior

Critical.

Conflict behavior

Critical when customer identity may be wrong.

Evidence expectation

Supporting label and page.

Acknowledgement

A legitimate difference from the customer record may be acknowledged with a reason after confirming customer identity. An unresolved document value cannot.

13.6 Mailing address**Expected behavior**

Capture the insured mailing address when clearly present.

Common invalid conditions

- missing address
- garaging address mistaken for mailing address
- agency address mistaken for insured address
- incomplete address
- conflict with customer contact address

Missing behavior

Important rather than critical. Reviewer may mark Not Present.

Conflict behavior

Important. The reviewer must confirm that the document still belongs to the customer.

Evidence expectation

Supporting page and surrounding label.

Acknowledgement

Allowed when the addresses are intentionally different.

13.7 Total policy premium**Expected behavior**

Capture only a clearly identified total policy premium.

Common invalid conditions

- several possible totals
- installment amount mistaken for total premium
- subtotal or vehicle premium mistaken for policy total
- missing currency context
- unreadable value

Missing behavior

Informational because the field is optional when not clearly present.

Conflict behavior

Important. The reviewer may select the correct total or mark Not Present.

Evidence expectation

Text should clearly connect the amount to total policy premium.

Acknowledgement

Allowed by marking Not Present. The product must not sum unrelated values automatically.

13.8 VIN

Expected behavior

Capture the VIN shown for each vehicle.

Common invalid conditions

- incomplete or truncated VIN
- OCR substitutions
- VIN associated with wrong vehicle
- duplicate VIN
- value is another identifier
- intentionally masked VIN

Missing behavior

Important when the vehicle is clearly listed but no VIN is available.

Conflict behavior

Critical when one VIN appears linked to several vehicles or several VINs appear for one vehicle.

Evidence expectation

Supporting page and association with the same vehicle.

Acknowledgement

A clearly masked or partial VIN may be approved after explicit important-warning acknowledgement. A malformed value presented as complete must be corrected or marked partial/not present.

13.9 Vehicle year

Expected behavior

A plausible model year associated with the vehicle.

Common invalid conditions

- non-numeric or impossible year
- year taken from policy date
- year conflicts with vehicle text
- year missing while make and model are present

Missing behavior

Important.

Conflict behavior

Important unless the conflict causes vehicle identity uncertainty, in which case it becomes critical.

Evidence expectation

Evidence from the vehicle listing.

Acknowledgement

Allowed when the document genuinely omits the year.

13.10 Vehicle make and model

Expected behavior

Readable make and model associated with the correct VIN and year.

Common invalid conditions

- make and model merged incorrectly
- trim mistaken for model
- vehicle values combined across rows
- missing make or model

Missing behavior

Important.

Conflict behavior

Critical when vehicle identities are mixed.

Evidence expectation

Same page or table row as the vehicle.

Acknowledgement

Allowed only when the document genuinely omits the value and vehicle identity remains understandable.

13.11 Driver name

Expected behavior

Each clearly listed driver appears as a separate person.

Common invalid conditions

- named insured duplicated as several drivers
- agent or household member mistaken for driver
- names merged
- duplicate driver
- missing drivers section

Missing behavior

Important when a driver section is visibly present but no names were extracted. Informational when the document appears not to list drivers.

Conflict behavior

Important.

Evidence expectation

Page and surrounding driver label.

Acknowledgement

Allowed when the document does not list drivers.

13.12 Bodily injury liability limit**Expected behavior**

Capture the liability limit in the form represented by the policy.

Common invalid conditions

- per-person and per-accident amounts reversed
- property-damage value included
- another coverage mistaken for bodily injury
- value lacks clear label
- several competing values

Missing behavior

Important.

Conflict behavior

Important or critical when coverage identity cannot be determined.

Evidence expectation

Coverage label and supporting page.

Acknowledgement

Allowed by marking Not Present when the document does not clearly show it. A conflicting selected value must be resolved before approval.

13.13 Property damage liability limit

Behavior follows the bodily injury limit rules.

The reviewer must be able to distinguish it from:

- premium
- deductible
- combined single limit
- uninsured motorist property damage

Missing is important, not automatically critical.

13.14 Uninsured or underinsured motorist limit

This is optional when clearly present.

Missing behavior

Informational.

Conflicting or weakly labelled behavior

Important.

Approval

Allowed when marked Not Present or when no supported value is available.

13.15 Comprehensive deductible

Expected behavior

Associate the deductible with the correct vehicle.

Common invalid conditions

- collision deductible mistaken for comprehensive
- deductible linked to wrong vehicle
- coverage marked as not included
- several deductible options
- amount mistaken for premium

Missing behavior

Informational when no comprehensive coverage is shown.

Conflict behavior

Important. Critical if association with the correct vehicle cannot be determined and a value would otherwise be approved.

Acknowledgement

Allowed when marked Not Present.

13.16 Collision deductible

Follows the same behavior as comprehensive deductible.

13.17 Expiration before effective date

This is a critical blocking conflict.

The reviewer must:

- correct one or both dates
- select the correct policy period
- or reject the document

Acknowledgement alone is not sufficient.

13.18 Duplicate vehicles

Possible duplicates are detected when vehicle information strongly overlaps.

The reviewer must:

- confirm they are separate vehicles
- merge by removing the incorrect duplicate
- or correct the identifiers

Approval is blocked when duplicate vehicle identity remains unresolved.

13.19 Duplicate drivers

Possible duplicates create an important warning.

The reviewer may:

- remove the duplicate
- confirm separate people with the same name
- correct the name

13.20 Incomplete VIN

The product distinguishes:

- clearly masked or partial VIN
- likely OCR damage
- malformed value presented as complete

Masked or partial VIN may be approved with acknowledgement.

Likely OCR damage should be corrected when source evidence is readable.

13.21 Conflicting policy numbers

Several plausible policy numbers are critical.

The reviewer must identify one correct policy number.

If one file appears to contain several policies, the reviewer should reject it and request a one-policy file.

13.22 Several possible premiums

Several possible totals create an important warning.

The reviewer may:

- select one clearly labelled total
- enter the supported total
- mark premium Not Present

The system must not calculate a total by adding values.

13.23 Coverage associated with the wrong vehicle

A vehicle-specific deductible associated with the wrong vehicle creates a critical association warning.

The reviewer must move or correct the value before approval.

13.24 Customer-record conflict

Name conflict

Critical.

Reviewer must confirm the document belongs to the selected customer.

Address conflict

Important.

Reviewer may retain both values and optionally update customer contact information during approval.

Email or phone conflict

The declaration-page extraction does not include email or phone, so no automatic conflict is created.

14. Approval workflow

14.1 When approval becomes available

Approval becomes available only when:

- the document is in Needs Review
- required policy identity fields are resolved
- all critical warnings are resolved
- important warnings are corrected, marked Not Present, or acknowledged
- no unsaved edits remain
- the review result is current
- the customer still exists and is accessible

14.2 Approval summary

Before confirmation, the reviewer sees:

- customer
- source document
- policy identity
- named insured
- dates
- premium when present
- vehicles
- drivers
- coverage
- reviewer corrections
- reviewer-added or removed items
- acknowledged warnings
- missing optional values
- effect on the current active policy
- optional contact-address update

14.3 Existing active policy

When no active policy exists:

- approval creates the first active policy version

When an active policy exists:

- the reviewer sees both current and proposed policy identity and dates
- the product recommends whether the proposed version should become current
- the reviewer confirms the result

14.4 Replacement recommendation

A proposed policy should normally become current when:

- it has a later effective period
- it represents a renewal or replacement
- the reviewer confirms it is the customer's current policy

An older declaration may be approved as historical without replacing the current policy.

14.5 Confirmation dialog

The final confirmation states:

- approval is a business action
- approved information will create a preserved policy version
- the active policy may change
- previous approved versions remain available
- AI output will not be treated as approved until this action completes

The user must explicitly select Approve.

14.6 Customer record changes

Successful approval:

- creates a reviewed policy version
- associates it with the customer
- marks it current or historical according to the reviewer's decision
- updates the customer's current-policy summary when applicable
- does not automatically overwrite staff-entered name, email, phone, or address
- updates contact address only when the reviewer explicitly selected that option

14.7 Repeated approval protection

If the user submits approval twice:

- only one approval result is created
- the second action returns the completed outcome
- no duplicate policy version or duplicate audit event is created

14.8 Approval failure

If approval cannot complete:

- the document remains Needs Review unless another user already completed it
- reviewer edits remain preserved
- the user sees that no policy change was confirmed
- the user can retry after refreshing current data
- email action remains unavailable

14.9 Customer changed during review

If customer information changed after the review page was opened:

- approval pauses
- the reviewer sees the latest customer information
- relevant differences are highlighted
- the reviewer must reconfirm customer association
- saved field corrections remain available

14.10 Processing result changed during review

If a newer processing result exists:

- approval is blocked
- the reviewer is told that the extraction changed
- the user must reload and review the latest result
- previous unsaved edits are not silently applied to the new result

14.11 Post-approval screen

After approval, show:

- Approved status
- active or historical policy designation
- approval time
- reviewer
- summary of final values
- corrections and acknowledged warnings
- links to Customer Details, Source Document, and Audit History
- Send Acknowledgement action when eligible

14.12 Audit events

Approval records:

- reviewer
 - time
 - source document
 - processing attempt
 - final approved values
 - material corrections
 - warnings acknowledged
 - whether the version became current
 - previous active policy when replaced
 - contact-address update when selected
-

15. Rejection workflow

15.1 When rejection is available

Rejection is available when:

- the document is Needs Review
- an approved document has a new unapproved reprocessing result
- a correction review has been opened but should not proceed

15.2 Required rejection reason

The reviewer must select one reason:

- Unsupported document type
- Unreadable document
- Wrong customer
- Several policies in one file
- Duplicate document
- Information cannot be confirmed
- Replacement document required
- Other

15.3 Optional notes

Notes are optional except for Other, where a short explanation is required.

15.4 Confirmation

The confirmation explains:

- the document will not update the customer policy
- the rejection remains in history
- a replacement document may be uploaded
- the result cannot be directly approved later without a new review result

15.5 Customer record effect

Rejection:

- does not create a policy version
- does not change the current active policy
- does not delete existing approved information

15.6 Document status

The document becomes Rejected.

15.7 Rejection reversal

Recommendation: Do not provide a direct Undo Rejection action.

Problem solved: Preserves a clear record of the original decision.

Simpler alternative: Change status back to Needs Review.

Why the simpler alternative is insufficient: It hides the fact that a reviewer previously rejected the result.

Accepted trade-off: Correcting a mistaken rejection requires a new review attempt or replacement upload.

15.8 Reprocessing rejected documents

Reprocessing may be available when:

- the rejection was caused by extraction quality
- the configuration changed
- the reviewer believes a new attempt may produce useful results

Reprocessing is not recommended for:

- wrong customer
- true unsupported document
- several policies in one file
- known duplicate

15.9 Replacement upload

A replacement can be uploaded from the rejection summary.

The rejected document remains preserved.

15.10 Audit event

The rejection event records:

- reviewer
- time
- reason
- optional note
- related customer
- source document
- current active policy remaining unchanged

16. Reprocessing and retry workflow

16.1 Action definitions

Release 1 distinguishes four actions.

Technical retry

Used after a temporary processing failure.

Examples:

- provider timeout
- temporary service error
- interrupted extraction
- delayed processing failure

Reprocessing

Used to intentionally run the original document through extraction again.

Examples:

- extraction instructions changed
- model behavior changed
- a previous result appears poor
- a benchmark comparison is needed

Correction review

Used when an approved result needs a human correction without requiring another paid AI attempt.

Replacement upload

Used when the source PDF itself is wrong, incomplete, corrupted, unreadable, or belongs to another customer.

16.2 Technical retry availability

Available only for retryable Failed results.

Not available for:

- corrupted PDF
- password-protected PDF
- page limit exceeded
- unsupported document
- wrong customer
- several policies in one file
- exhausted retry allowance

16.3 Attempt limits

Recommendation: Allow no more than two user-triggered technical retries after the initial attempt.

Problem solved: Prevents repeated paid calls and unproductive loops.

Simpler alternative: Unlimited retry.

Why the simpler alternative is insufficient: It creates cost risk and hides persistent failures.

Accepted trade-off: Some documents will require replacement or later investigation after the limit is reached.

16.4 Paid-processing warning

Before a retry or reprocessing action that may use paid services, show:

- that another processing attempt will occur
- whether the environment permits live processing
- current retry count
- whether OCR may run again
- whether previous results remain preserved

The user must confirm.

16.5 Preservation of prior attempts

Every attempt remains visible with:

- attempt number or chronological order
- start and completion time
- processing method
- result
- failure category
- model or extraction version identifier in user-understandable form
- whether it was live or replayed

16.6 Reviewer corrections

Technical retry or reprocessing does not erase reviewer corrections.

Corrections remain attached to the previous review result or approved version.

The new result starts as a separate review candidate.

16.7 Approved customer data

An active approved policy remains unchanged during:

- retry
- reprocessing
- correction review

Only another approval can create or replace an approved version.

16.8 Resulting status

Failed document retry

Failed becomes Processing.

If successful, it becomes Needs Review.

If unsuccessful, it returns to Failed with the new attempt visible.

Approved document reprocessing

The document retains its approved-history context and shows reprocessing activity.

When successful, a new review result becomes available.

The current approved policy remains active.

16.9 Duplicate protection

Starting the same retry or reprocessing action twice:

- creates only one active attempt
- returns the existing active attempt status
- does not consume duplicate usage

16.10 Uploading a corrected document

A corrected source file is uploaded as a new document.

The user may indicate that it replaces or corrects a previous failed or rejected document.

The original remains in history.

17. Approved policy versions

17.1 Version principle

Recommendation: Approved policy versions are preserved and cannot be silently edited.

Problem solved: Users need to know exactly what was approved, from which document, by whom, and when.

Simpler alternative: Maintain one editable policy record.

Why the simpler alternative is insufficient: It destroys historical traceability and makes later corrections difficult to explain.

Accepted trade-off: Corrections create a new approved version instead of changing the old one.

17.2 Current active policy

A customer may have one current active personal auto policy version in Release 1.

The current version is the one staff considers operationally current.

17.3 Previous versions

Every previously approved version remains available.

Each version shows:

- current or historical designation
- carrier
- policy number
- effective and expiration dates
- source document
- reviewer
- approval date
- replacement relationship

17.4 Source relationship

Every approved version links to:

- source PDF
- review result
- processing attempt
- reviewer corrections
- audit history

17.5 Newer document

When a newer declaration is approved:

- it normally becomes current
- the previous version becomes historical
- the previous version remains unchanged
- the replacement is recorded

17.6 Older document

When an older declaration is approved:

- the product recommends keeping the existing newer policy current
- the reviewer may approve the older version as historical
- making an older version current requires explicit confirmation

17.7 Manual deactivation

Recommendation: Do not include a general Deactivate Policy action in Release 1.

Problem solved: There is no approved business workflow for policy cancellation, non-renewal, or carrier verification.

Simpler alternative: Allow users to mark policies inactive.

Why the simpler alternative is insufficient: It could imply policy status knowledge that the product does not have.

Accepted trade-off: “Current” means the agency’s selected current uploaded version, not verified carrier status.

17.8 Direct editing

Approved values cannot be edited directly.

To correct an approved version:

1. Start Correction Review.
2. Review the existing approved values against the source.
3. Make corrections.
4. Approve a new corrected version.
5. Preserve the original approved version as historical.

17.9 Version comparison

Release 1 should provide a simple comparison when replacing an active version.

Show changed values for:

- carrier
- policy number
- dates
- premium
- named insured
- mailing address
- vehicles
- drivers
- coverage

It does not need a general side-by-side comparison tool for any two arbitrary versions.

18. Audit history

18.1 Purpose

Audit history explains business-relevant activity in language staff can understand.

It is not a technical diagnostic log.

18.2 Events included

Audit history includes:

- customer created
- customer contact information changed
- document uploaded
- duplicate warning shown
- intentional duplicate confirmed
- processing started
- direct text used
- OCR used
- processing completed
- processing failed
- retry started and completed
- reprocessing started and completed
- review opened when useful
- field corrected
- vehicle or driver added
- vehicle or driver removed
- field marked Not Present
- warning acknowledged
- document approved
- policy version created
- active policy replaced
- document rejected
- acknowledgement email requested
- acknowledgement email sent
- acknowledgement email failed
- demo data reset

18.3 Event wording

Events should use clear wording such as:

- "Jordan Lee uploaded auto-declaration.pdf for Maria Santos."
- "OCR was used because the PDF did not contain enough readable text."
- "Jordan Lee changed the policy expiration date from June 1 to July 1."
- "Jordan Lee approved policy version 2 as the current policy."
- "Acknowledgement email could not be sent. Policy approval was not affected."

18.4 Actor

Each user action shows:

- actor name
- demo-user label when applicable
- system label for automated processing events

18.5 Timestamp

Show:

- clear local date and time
- exact timestamp on detail or hover where useful
- consistent timezone behavior

18.6 Related objects

Each event links to the related:

- customer
- document
- review
- approved policy version

when applicable.

18.7 Previous and new values

For material reviewer corrections, show:

- field name
- previous candidate or approved value
- new value
- actor
- time

Sensitive values should not be unnecessarily repeated in broad activity lists. Detailed value changes may appear in the document audit view.

18.8 Processing attempts

Each attempt shows:

- attempt order
- trigger type
- live or replayed
- processing method
- completion or failure
- failure category
- user who initiated a retry or reprocessing action

18.9 Warning acknowledgements

Show:

- warning
- reviewer
- acknowledgement reason when required
- time

18.10 Filtering

Release 1 audit filtering should be limited to:

- event category
- actor
- date range
- related document

18.11 Pagination

Use page-based pagination or a “Load More” control.

The default shows newest events first.

18.12 Empty state

When no audit activity exists:

- state that no recorded activity exists yet
- do not display a blank timeline without explanation

18.13 Technical logs distinction

Normal users should not see:

- raw stack traces
- internal service names
- secret values
- complete model prompts
- complete provider responses
- low-level network errors

A user-facing failure reason and reference identifier are sufficient.

19. Acknowledgement email workflow

19.1 Availability

The Send Acknowledgement action is available when:

- a document has been approved
- the related customer has a valid email address
- the environment permits sending
- the user has not exceeded email restrictions
- no send is currently in progress

19.2 Default recipient

The default recipient is the customer’s staff-entered email address.

The declaration-page extraction does not provide an email address.

19.3 Recipient restrictions

Normal staff users may:

- use the customer email
- correct the customer email through Edit Customer before sending

Release 1 should not allow arbitrary additional recipients, CC, or BCC.

Demo users may send only to:

- a configured safe test recipient
- or a simulated delivery destination

19.4 Message purpose

The fixed message acknowledges that:

- the agency received the document
- the document was reviewed
- the agency may follow up if more information is required

19.5 Message restrictions

The email must not claim:

- the policy is currently active
- the carrier verified the policy
- coverage is sufficient
- the customer is fully insured
- the agency has bound coverage
- the premium is final
- the customer should cancel another policy

19.6 Preview

Before sending, show:

- recipient
- fixed subject
- fixed message body
- related customer
- related approved document

The user may not redesign the template.

19.7 Confirmation

The user confirms Send.

The product explains that email success does not affect policy approval.

19.8 Sending state

While sending:

- disable repeated Send actions
- show a clear sending indicator
- allow the user to remain on the page

19.9 Success

On success:

- show a temporary success message
- mark the email as sent in the approved summary
- record an audit event
- retain send time and recipient

19.10 Failure

On failure:

- policy approval remains unchanged
- show a persistent failure message on the approved summary
- allow a limited resend
- preserve the failed attempt in audit history

19.11 Duplicate send warning

If an acknowledgement was already sent:

- show recipient and send time
- replace the primary action with Send Again
- require confirmation
- explain that the customer may receive a duplicate message

19.12 Demo behavior

Demo mode:

- clearly labels email as simulated or restricted
 - does not send to arbitrary customer addresses
 - may show a successful simulated send for Loom demonstrations
 - records a demo audit event
-

20. Usage and cost controls

20.1 Usage view decision

Recommendation: Include a simple operational Usage screen.

Problem solved: Users and the product owner need to understand whether live document processing is available and which activities create cost.

Simpler alternative: Hide all usage information.

Why the simpler alternative is insufficient: Cost control is a core Release 1 requirement, especially for public demos.

Accepted trade-off: The screen reports operational usage without becoming a billing or finance product.

20.2 Information shown

Show for the current configured period:

- documents submitted
- documents processed live
- documents processed using stored results
- pages processed
- pages sent through OCR
- AI extraction attempts
- technical retries
- intentional reprocessing attempts
- failed attempts
- estimated processing cost when sufficiently reliable
- current processing limit
- remaining allowance
- whether live processing is enabled

20.3 Estimated cost

Estimated cost must be labelled as an estimate.

When cost cannot be calculated reliably:

- show usage quantities
- do not invent a monetary value
- explain that provider billing may differ

20.4 Limit status

Possible limit states:

- Available
- Approaching Limit

- Limit Reached
- Live Processing Disabled

20.5 Approaching limit

When approaching the limit:

- show a persistent warning on Usage
- show a smaller warning before paid processing
- existing processing continues
- stored results remain available

20.6 Limit reached

When the limit is reached:

- new live processing is disabled
- existing stored documents and results remain viewable
- review, approval, rejection, audit, and demo replay remain available
- upload may be disabled or limited to samples that use stored results
- the user sees when the limit period resets or that an authorized configuration change is required

20.7 Who can trigger live processing

Normal staff users may trigger live processing within configured limits.

Controlled demo users may trigger live processing only when:

- the demo specifically permits it
- the sample is approved
- the remaining demo allowance is sufficient

20.8 Replay identification

A replayed result is clearly marked:

- Sample Result
- Stored Processing Result
- No live AI processing used

It should not pretend to be a newly generated live result.

20.9 Paid-processing warning

Show a warning before:

- intentional duplicate processing
- technical retry
- reprocessing
- optional demo live run

A normal first upload does not need an excessive confirmation if current limits are healthy and the file is within scope.

20.10 Retry-limit behavior

When retry allowance is exhausted:

- retry action is disabled
 - the reason is displayed
 - upload replacement remains available when appropriate
 - previous attempts remain viewable
-

21. Public demonstration mode

21.1 Demo objective

Demo mode must support:

- Loom recording
- live portfolio walkthrough
- client demonstration
- interview explanation
- safe exploration by a visitor

without requiring developer access.

21.2 Demo entry

The sign-in screen provides Enter Demo when enabled.

Entering demo mode:

- creates or opens a controlled synthetic-data session
- clearly marks the environment as Demo
- displays a short disclaimer
- provides guided sample actions

21.3 Synthetic data

All demo customers, documents, policy numbers, names, addresses, vehicles, and emails must be synthetic.

No real customer information may appear.

21.4 Sample declaration pages

Demo mode includes a small curated set:

- clean text PDF

- readable scanned PDF
- document with warnings
- failed or unsupported example
- approved historical example

21.5 Stored processing results

Most sample workflows use stored results.

The interface identifies:

- that the result is replayed
- which processing stages are being demonstrated
- that no live AI cost was incurred

21.6 Limited live processing

When enabled:

- only approved sample documents may use live processing
- strict session and period limits apply
- the action requires confirmation
- current allowance is visible
- arbitrary public uploads remain disabled or tightly restricted

21.7 Upload restrictions

Recommendation: Controlled demo users should not upload arbitrary local documents in the public demo.

Problem solved: Prevents sensitive information exposure, unsupported content, storage abuse, and uncontrolled AI cost.

Simpler alternative: Allow unrestricted uploads with a file-size limit.

Why the simpler alternative is insufficient: It does not control privacy, repeated processing, or inappropriate files.

Accepted trade-off: Visitors cannot test their own document, but the demo remains safe and repeatable.

A private owner-controlled demonstration may permit a manually authorized live upload outside public demo mode.

21.8 Reset behavior

Demo mode provides Reset Demo.

Reset:

- asks for confirmation

- restores the synthetic starting state
- removes disposable demo changes
- preserves no real information
- returns the user to Overview
- records a demo reset event only within the current demo context

21.9 Protected actions

Demo users cannot:

- delete core sample records permanently
- change organization settings
- send email to arbitrary recipients
- exceed live-processing limits
- access real customer information
- alter provider or cost settings
- create unlimited customers or uploads
- perform unrestricted destructive actions

21.10 Session isolation

Recommendation: Demo changes should be isolated to the current demo session where practical.

Problem solved: One visitor should not break another visitor's walkthrough or the owner's Loom recording.

Simpler alternative: Use one shared mutable demo workspace.

Why the simpler alternative is insufficient: Concurrent users can alter statuses and remove predictable sample conditions.

Accepted trade-off: Session isolation adds later architectural responsibility but significantly improves demo reliability.

21.11 Email restrictions

Demo email is:

- simulated
- or sent only to a controlled safe destination

The interface clearly identifies which behavior occurred.

21.12 Provider outage

During a provider outage:

- replayed samples remain fully usable
- live processing is disabled with an explanation
- the demo still supports upload simulation, processing-state walkthrough, review, approval, policy history, audit, and email simulation

21.13 Visible disclaimer

The demo states:

- data is synthetic
- results are for demonstration
- the product does not verify active insurance coverage
- AI results require human review
- no insurance advice is provided

21.14 User guidance

Demo Overview should offer a small guided path:

1. Open sample customer.
2. Process or replay a declaration page.
3. Review warnings.
4. Approve the result.
5. View policy history and audit.
6. Simulate acknowledgement email.
7. Reset demo.

This is guidance, not a separate tutorial product.

22. Notifications and user feedback

22.1 Inline validation

Use inline validation for:

- customer form fields
- review field formats
- email format
- rejection reason
- required acknowledgement reason
- upload file requirements

The message should identify:

- what is wrong
- what the user must change

22.2 Page-level alerts

Use page-level alerts for:

- session expiration
- stale review result
- customer changed during review
- provider outage

- processing limit reached
- source document unavailable
- demo restrictions
- existing approved policy remaining active during reprocessing

These alerts remain until the condition is resolved or the user leaves.

22.3 Temporary success messages

Use temporary success messages for:

- customer saved
- upload accepted
- review edit saved
- document approved
- document rejected
- email sent
- filters cleared
- demo reset completed

The page itself must also reflect the successful result.

22.4 Temporary error messages

Use temporary errors for simple recoverable failures, such as:

- local action could not complete
- temporary list refresh failed
- email send failed while the page already shows persistent status

Critical failures should also appear in the page content.

22.5 Confirmation dialogs

Use confirmation for:

- approval
- rejection
- intentional duplicate processing
- retry or reprocessing with possible paid usage
- removing a vehicle
- removing a driver
- replacing the current active policy
- sending a duplicate acknowledgement
- resetting demo
- discarding unsaved changes

Do not use confirmation for harmless navigation or ordinary field edits.

22.6 Persistent status banners

Use persistent banners for:

- Processing
- Taking longer than expected
- Failed
- Needs Review with unresolved critical warnings
- Approved policy remains active during reprocessing
- Demo mode
- Live processing disabled
- stale data

22.7 Background processing notifications

When processing completes while the user is elsewhere in the product:

- show one in-product notification
- update relevant Overview and queue counts
- provide a direct action to open the review or failure
- do not repeatedly notify for the same completion

Release 1 does not require browser push notifications, SMS, or email notifications for processing completion.

23. Loading, empty, disabled, and failure states

23.1 Sign-in

Loading

- sign-in action shows progress
- repeated submission disabled

Empty

- credential fields empty
- demo option shown when available

Failure

- invalid credentials or unavailable sign-in shown clearly
- retry allowed

Session failure

- explain expiration and return path

23.2 Overview

Loading

- operational placeholders

Empty

- zero counts and onboarding actions

Partial data

- unavailable section shows local error

Failure

- retry page summary

Disabled

- Upload action disabled when no customer exists or processing limit prevents upload

23.3 Customer list

Loading

- placeholder rows

Empty

- Create Customer action

No results

- Clear Filters

Partial data

- missing policy summary shown as unavailable, not as zero

Failure

- list retry

Session failure

- sign-in and return

23.4 Customer detail

Loading

- summary and section placeholders

Empty

- no documents
- no approved policy
- no activity

Partial data

- available customer information remains visible if one section fails

Failure

- retry section or page

Stale state

- banner when customer changed elsewhere

23.5 Document upload

Loading

- customer search or file inspection progress

Empty

- no file selected

Disabled

- submit disabled without customer and valid file
- disabled when live processing unavailable and no replay path exists

Validation failure

- unsupported type, size, page count, encryption, corruption

Interrupted upload

- retry action

Session failure

- explain whether upload was accepted

23.6 Documents list

Loading

- placeholder rows

Empty

- Upload Document

No results

- clear filters

Failed-list state

- failure categories and recovery actions

Partial data

- missing warning summary labelled unavailable

Failure

- retry list

23.7 Review Queue

Loading

- queue placeholders

Empty

- queue clear message

No results

- clear filters

Stale item

- completed status and redirect

Failure

- retry queue

23.8 Review workspace

Loading

- field and document placeholders independently

Partial data

- source viewer may fail while fields remain visible
- evidence may be unavailable for specific fields

Disabled

- approval disabled with listed reasons
- edits disabled after another user completes review
- live reprocessing disabled at usage limit

Provider failure

- appears through failed attempt or reprocessing notice

Stale data

- final actions disabled until reload

Session failure

- reauthentication and unsaved-change handling

23.9 Usage view

Loading

- usage placeholders

Empty

- zero usage with explanation

Partial data

- usage counts shown even if cost estimate unavailable

Failure

- retry

Limit reached

- persistent disabled-processing state

23.10 Audit history

Loading

- timeline placeholders

Empty

- no recorded activity

No results

- clear filters

Partial data

- event shown without optional detail rather than hidden

Failure

- retry activity

23.11 Unknown failure principle

An unknown failure must:

- state that the action did not complete as expected
 - avoid claiming data was lost
 - show a retry or safe next action
 - provide a reference identifier where useful
 - never expose raw internal errors
-

24. Search, filtering, sorting, and pagination

24.1 General principles

- Search and filters should be simple and visible.
- Active filters should be clearly displayed.
- Clear All should restore the default view.
- Search, filter, sort, and page should remain in the page address where useful.
- No saved views or advanced query language is required.

24.2 Customers

Searchable

- name
- email
- phone
- current policy number
- current named insured

Filters

- needs review
- failed documents
- active policy
- no active policy

Default sort

- recent activity first

Pagination

- page-based

24.3 Documents

Searchable

- file name
- customer
- carrier
- policy number

Filters

- status
- warning severity
- processing method
- date range
- duplicate
- live or replayed

Default sort

- newest upload first

Pagination

- page-based

24.4 Review Queue

Searchable

- customer
- document
- carrier
- policy number

Filters

- warning severity
- OCR
- existing approved policy
- age

Default sort

- critical warnings first, then oldest

Pagination

- page-based when needed

24.5 Audit history

Search

A general text search is not required in Release 1.

Filters

- event category
- actor
- date
- document

Default sort

- newest first

Pagination

- Load More or page-based pagination

24.6 Empty filtered result

Every filtered screen should:

- state that no records match
 - show active conditions
 - provide Clear Filters
 - preserve the user's broader data
-

25. Concurrency and stale-data behavior

25.1 Two users open the same document

Both may view the document.

The product should not silently let one user overwrite another user's completed work.

Before saving material edits, approving, or rejecting:

- current status is checked
- the user is warned if the review changed
- stale final actions are blocked

25.2 One user approves while another reviews

The second user sees:

- a persistent message that the document was approved by another user
- final actions disabled

- the page converted to read-only approved summary after refresh
- their unsaved changes are not applied

25.3 One user rejects while another reviews

The same stale behavior applies.

The second user may inspect the rejection result but cannot continue the previous review.

25.4 Customer edited during review

The reviewer may continue field review.

Before approval:

- latest customer information is displayed
- name or address changes are compared
- customer association must be reconfirmed when relevant

25.5 Processing completes while page is open

The page updates status without requiring the user to restart the workflow.

If automatic update is unavailable:

- a clear Refresh Result action appears
- the page must not remain indefinitely on an outdated stage without explanation

25.6 Retry completes after user leaves

Documents, Overview, and Review Queue reflect the completed status when the user returns.

An in-product notification may link to the result.

25.7 Same action submitted twice

For upload, approval, rejection, retry, reprocessing, and email:

- the product treats repeated identical submission safely
- the user receives the existing result
- no duplicate business outcome occurs

25.8 Outdated page

When important data is outdated:

- show a stale-data banner
- disable final actions
- require refresh
- preserve unsaved edits visually where practical until the user decides how to proceed

25.9 Session expires during critical action

After reauthentication:

- show the actual current status
- do not assume the action succeeded or failed
- return the user to the relevant page
- prevent duplicate completion

25.10 Concurrency decision

Recommendation: Use stale-data detection and safe final-action checks rather than user assignment or hard review locks.

Problem solved: Prevents overwrites while keeping Release 1 simple.

Simpler alternative: Allow last save to win.

Why the simpler alternative is insufficient: It can erase reviewer work or create contradictory approval outcomes.

More complex alternative: Assign and lock every review.

Why it is unnecessary: Release 1 does not need workload routing or formal ownership.

Accepted trade-off: Two users may occasionally need to reload and repeat a small amount of work.

26. Responsive and accessibility behavior

26.1 Desktop-first expectation

The full review experience is designed primarily for laptop and desktop screens.

The preferred layout should support:

- readable PDF
- visible field panel
- accessible review actions
- comparison without excessive switching

26.2 Minimum screen behavior

At narrower desktop widths:

- panels may resize
- source and review areas remain readable
- actions must not overlap content
- horizontal scrolling of the entire application should be avoided

26.3 Tablet and narrow screens

Use a source/data tab or stacked layout.

The user can:

- review fields
- open evidence
- edit values
- approve when the device has enough usable space

26.4 Mobile limitations

Mobile should support:

- sign-in
- Overview
- customer lookup
- status viewing
- activity viewing
- simplified document preview

Full review editing and approval may be intentionally unavailable.

The limitation must be stated clearly rather than presenting a broken layout.

26.5 Keyboard navigation

Core actions should be reachable by keyboard:

- navigation
- search
- forms
- field editing
- warning acknowledgement
- page controls
- approval and rejection dialogs

Keyboard order should follow the visible workflow.

26.6 Visible focus

Every interactive element must show a clear focus state.

26.7 Form labels

Every field must have a persistent text label.

Placeholder text alone is insufficient.

26.8 Error identification

Errors must identify:

- the affected field or action
- the problem
- the required correction

Error meaning must not depend only on color.

26.9 Warning meaning

Warnings should use:

- severity label
- icon or shape
- explanatory text
- color as an additional signal only

26.10 Contrast and readability

Text, controls, warnings, disabled states, and focus indicators must remain readable under normal display conditions.

26.11 Document viewer limitations

PDF viewers have accessibility limitations.

Release 1 should reduce the impact by providing:

- extracted evidence text
- page numbers
- field labels
- keyboard-accessible viewer controls
- downloadable synthetic or authorized source files
- clear warnings when the visual document cannot be fully accessed

The product should not claim that the PDF viewer itself makes every source document accessible.

27. Functional acceptance criteria

27.1 Authentication

- A valid staff user can sign in and reach Overview.
- Invalid credentials do not reveal whether an account exists.
- An unauthenticated user cannot open protected customer or document pages.
- Sign-out prevents further protected actions.
- An expired session returns the user to the same protected page after successful sign-in where possible.
- A repeated sign-in submission does not create several sessions or navigation results.

- Demo entry creates a visibly restricted demo context.

27.2 Customers

- A customer can be created with full name only.
- Invalid email prevents save.
- Duplicate candidates are shown before a likely duplicate customer is created.
- A user may intentionally continue after reviewing duplicate candidates.
- Editing contact information does not alter approved policy history.
- Customer list search finds matching name, email, phone, or policy number.
- Customer Detail distinguishes contact information from approved policy information.
- A customer with no approved policy shows an explicit empty state.

27.3 Uploads

- A supported PDF can be uploaded for a selected customer.
- A non-PDF file is rejected before processing.
- A PDF over 15 MB is rejected.
- A PDF over 10 pages is rejected.
- A password-protected PDF is rejected with an unlock instruction.
- A corrupted PDF does not enter normal processing.
- Cancelling before upload acceptance creates no processing document.
- An interrupted upload can be retried safely.
- Multiple-file selection results in one-file guidance.
- A duplicate warning appears before another paid attempt.
- Intentional duplicate processing requires confirmation.
- Successful upload enters Processing and appears in Documents.

27.4 Processing

- Processing continues when the user leaves the page.
- Processing shows a meaningful current description.
- Direct text and OCR usage are distinguishable.
- Successful processing enters Needs Review.
- A provider timeout eventually produces a clear retryable failure when recovery does not succeed.
- Unsupported documents do not produce an approvable result.
- Invalid extraction structure does not appear as a successful review.
- Delayed processing shows a taking-longer-than-expected message.
- A completed background result updates the product without requiring a new upload.

27.5 Review Queue

- Every Needs Review document appears in the queue.
- Failed documents do not appear in the queue.
- Critical-warning documents appear before lower-risk documents by default.
- Queue search and filters produce predictable results.
- Opening an item leads to the review workspace.
- An item disappears after approval or rejection.
- Opening an already completed stale item shows the completed result.

27.6 Review Workspace

- The reviewer can view the source and extracted values in one workflow.

- Selecting evidence opens the supporting page when available.
- Every field shows its current review state.
- Reviewer edits preserve the original AI candidate.
- Invalid edited values cannot be saved.
- A field can be marked Not Present where allowed.
- A missed vehicle or driver can be added.
- An incorrect vehicle or driver can be removed.
- Unsavd changes trigger a navigation warning.
- Another user's completed review cannot be overwritten.
- PDF rendering failure produces a visible review limitation.
- Approval readiness accurately reflects unresolved warnings.

27.7 Validation

- Missing carrier, policy number, policy dates, or primary named insured blocks approval.
- Expiration before effective date blocks approval.
- Conflicting policy numbers block approval.
- Customer-name conflict requires confirmation.
- Address conflict requires review but may be acknowledged.
- Missing optional premium does not block approval.
- Several possible premiums require resolution or Not Present.
- Duplicate vehicles require resolution.
- Missing optional uninsured-motorist coverage does not block approval.
- Vehicle-specific deductibles cannot remain associated with an unresolved wrong vehicle.
- Model self-confidence alone never marks a value approved.

27.8 Approval

- Approval is unavailable before required review conditions are complete.
- Approval displays final values and acknowledged warnings.
- Approval creates a preserved reviewed policy version.
- An existing active policy is not replaced without confirmation.
- Repeated approval creates only one approved outcome.
- Approval does not automatically change staff-entered contact information.
- A selected contact-address update is explicit and audited.
- Approval failure does not falsely show an approved status.
- Email action becomes available only after successful approval.

27.9 Rejection

- Rejection requires a reason.
- Other requires a note.
- Rejection does not alter the active customer policy.
- Rejection creates a preserved final outcome.
- Rejected results cannot be directly changed back to Needs Review.
- Replacement upload remains available.
- Reprocessing is available only when appropriate.

27.10 Retries and reprocessing

- Retry is available only for retryable technical failures.
- No more than two user-triggered technical retries are allowed.

- Paid-processing warning appears before retry or reprocessing.
- Previous attempts remain visible.
- Reviewer corrections are not erased.
- Active approved policy data remains unchanged.
- Repeated retry submission creates one active attempt.
- Failed reprocessing does not invalidate the existing approved policy.

27.11 Policy versions

- One version is clearly marked current.
- Previous approved versions remain visible.
- Every version links to its source document and reviewer.
- A newer approval can replace the current version.
- An older approval can be stored as historical.
- Approved values cannot be directly edited.
- Correction Review produces a new version.
- No approved version is silently rewritten.

27.12 Audit history

- Customer creation appears in activity.
- Upload, processing, OCR, retry, review, approval, rejection, and email events appear.
- Material corrections show previous and new values.
- Each user action identifies the actor and time.
- User-facing history does not expose raw technical errors.
- Filters narrow events without deleting them.
- Duplicate confirmations and warning acknowledgements are recorded.

27.13 Email

- Email is unavailable without an approved document and valid customer email.
- Preview shows the fixed message and recipient.
- The user cannot add arbitrary recipients.
- A successful send is recorded.
- A failed send does not reverse approval.
- Sending twice requires duplicate confirmation.
- Demo mode cannot send to arbitrary addresses.
- The message does not claim policy verification or coverage sufficiency.

27.14 Usage limits

- Usage shows documents, pages, OCR, AI attempts, failures, and replayed results.
- Estimated cost is labelled as estimated.
- No cost is invented when unavailable.
- Approaching-limit warning appears before the limit is reached.
- Limit reached disables new live processing.
- Existing results remain viewable and reviewable.
- Replay mode remains available.
- Disabled actions explain why they are disabled.

27.15 Demo mode

- Demo entry uses synthetic data only.

- Demo mode is visibly labelled.
 - Stored sample results work during provider outage.
 - Arbitrary public uploads are unavailable.
 - Live processing is limited and confirmed.
 - Email is simulated or restricted.
 - Reset restores the starting state without developer help.
 - Demo users cannot permanently destroy core samples.
 - Separate demo sessions do not expose one another's changes where session isolation is enabled.
 - The product displays an AI and insurance-verification disclaimer.
-

28. End-to-end scenarios

28.1 Clean text PDF successfully approved

Starting state

- Staff user is signed in.
- Customer exists.
- Processing allowance is available.
- Customer has no active policy.

User actions

1. User opens Customer Details.
2. User selects Upload Document.
3. User selects a valid clean text PDF.
4. User submits.
5. User opens the completed review.
6. User verifies evidence and values.
7. User approves.

System behavior

- File passes validation.
- Duplicate check finds no match.
- Document enters Processing.
- Direct text extraction is used.
- OCR is not used.
- Candidate fields and evidence are created.
- Validation finds no blocking conflict.
- Document enters Needs Review.
- Approval creates the first current policy version.

Visible statuses

Processing → Needs Review → Approved.

Recovery

No recovery required.

Final state

- Customer has a current approved policy.
- Source document remains linked.
- Acknowledgement email is available.

Audit history

- upload
- processing started
- direct text used
- processing completed
- review corrections if any
- approval
- policy version created

28.2 Readable scanned PDF approved after corrections

Starting state

Customer exists with no current policy.

User actions

1. Upload scanned PDF.
2. Open review after processing.
3. Correct OCR-damaged policy number.
4. Correct one VIN character.
5. Acknowledge OCR-quality warning.
6. Approve.

System behavior

- Direct text is insufficient.
- OCR is used.
- Candidate values are produced.
- Policy number and VIN receive warnings.
- Original AI values remain preserved after edits.
- Approval becomes available after correction and acknowledgement.

Visible statuses

Processing with "Reading scanned pages" → Needs Review → Approved.

Recovery

Reviewer corrects values using the source.

Final state

Approved policy contains reviewer-corrected values.

Audit history

- OCR used
- field corrections with previous and new values
- warning acknowledged
- approval

28.3 Missing optional fields

Starting state

Customer exists.

User actions

1. Upload declaration without clear total premium or uninsured-motorist limit.
2. Review extracted fields.
3. Mark premium Not Present if the system raised an uncertain candidate.
4. Approve.

System behavior

- Missing optional values are informational.
- Approval remains available once required fields are complete.
- No values are invented.

Visible statuses

Processing → Needs Review → Approved.

Recovery

No technical recovery.

Final state

Approved version contains explicit missing optional values.

Audit history

- processing
- Not Present action if used
- approval

28.4 Critical field conflict

Starting state

A document contains two different policy numbers.

User actions

1. Open review.
2. Inspect both evidence excerpts.
3. Select or enter the correct policy number.
4. Approve.

System behavior

- Critical warning blocks approval.
- Evidence points to both candidates.
- Correction resolves the warning.

Visible statuses

Needs Review remains until corrected, then Approved.

Recovery

Reviewer resolves the conflict or rejects the document if it contains several policies.

Final state

One supported policy number is approved.

Audit history

- conflict warning
- correction
- approval or rejection

28.5 Unsupported document

Starting state

User uploads a home insurance declaration or unrelated document.

User actions

1. Submit PDF.
2. Open failure summary.

System behavior

- File validation passes as PDF.
- Document-type evaluation identifies unsupported content.
- No approvable result is created.

Visible statuses

Processing → Failed, reason Unsupported Document.

Recovery

Upload a supported personal auto declaration.

Final state

Customer policy remains unchanged.

Audit history

- upload
- processing
- unsupported-document failure

28.6 Corrupted PDF

Starting state

User selects a damaged PDF.

User actions

1. Select file.
2. Attempt upload.

System behavior

- File cannot be opened.
- When detected before acceptance, submission is rejected.
- When detected after upload, document becomes Failed.

Visible statuses

Either no document status, or Processing briefly → Failed.

Recovery

Upload another copy.

Final state

No policy result created.

Audit history

Upload failure is recorded only if a lasting document was created.

28.7 Duplicate upload

Starting state

The same PDF was previously approved for the customer.

User actions

1. Select the same file.
2. Review duplicate warning.
3. Open existing document or intentionally continue.

System behavior

- New paid processing is paused.
- Existing document details are shown.
- Continuing requires confirmation.

Visible statuses

The new upload remains pending confirmation. If cancelled, no new processing occurs. If confirmed, Processing begins.

Recovery

Open existing approved result.

Final state

Normally no duplicate processing occurs.

Audit history

- duplicate detected
- user cancellation or intentional override

28.8 AI provider timeout

Starting state

Valid PDF has completed text extraction.

User actions

1. Upload document.
2. Later open failure summary.
3. Select Technical Retry.

System behavior

- Initial provider recovery does not complete.
- Document becomes Failed.
- Retry warning explains possible paid usage.
- Retry creates another attempt.

Visible statuses

Processing → Failed → Processing → Needs Review or Failed.

Recovery

Up to two user-triggered retries.

Final state

Either reviewable result or preserved failure.

Audit history

- timeout failure
- retry initiated
- retry outcome

28.9 OCR failure**Starting state**

Scanned PDF requires OCR.

User actions

1. Upload.
2. Open failure.
3. Upload clearer replacement.

System behavior

- Direct text is insufficient.
- OCR fails or produces no usable text.
- AI extraction does not proceed as a normal success.

Visible statuses

Processing → Failed.

Recovery

Technical retry if failure was temporary, otherwise replacement upload.

Final state

Original failed document remains; replacement processes separately.

Audit history

- OCR attempted
- OCR failed
- retry or replacement relationship

28.10 Extraction structure failure

Starting state

Usable text exists, but AI result cannot be interpreted as the required field structure.

User actions

1. Open failed result.
2. Retry if allowed.

System behavior

- Controlled internal handling fails to produce a valid result.
- No partial candidate is treated as reviewable.
- Document becomes Failed.

Visible statuses

Processing → Failed.

Recovery

Technical retry or later reprocessing.

Final state

No customer policy change.

Audit history

- extraction attempted
- invalid-result failure
- retry when used

28.11 Retry after technical failure

Starting state

Document failed because of temporary processing error.

User actions

1. Select Retry Technical Processing.
2. Confirm possible usage.

3. Leave page.
4. Return after notification.

System behavior

- One active retry starts.
- Previous failure remains visible.
- Retry succeeds and enters Needs Review.

Visible statuses

Failed → Processing → Needs Review.

Recovery

Review and approve normally.

Final state

A reviewable result exists.

Audit history

- retry initiated
- processing attempt
- successful completion

28.12 Reprocessing without overwriting an approved policy

Starting state

Customer has an active approved policy from the document.

User actions

1. Open approved result.
2. Select Reprocess.
3. Confirm cost warning.
4. Review new result.

System behavior

- Existing approved policy remains active.
- New attempt processes independently.
- New result enters Needs Review.
- No customer policy changes until another approval.

Visible statuses

Approved with Reprocessing indicator → New Review Available.

Recovery

Reviewer may approve, reject, or leave the existing approval unchanged.

Final state

Existing policy remains current unless the new result is approved.

Audit history

- reprocessing initiated
- new attempt
- result
- later approval or rejection

28.13 Newer declaration replaces current active policy

Starting state

Customer has current policy ending soon.

User actions

1. Upload newer declaration.
2. Review and approve.
3. Confirm that it becomes current.

System behavior

- Approval summary compares current and proposed dates.
- New version becomes current.
- Previous version becomes historical.

Visible statuses

New document: Processing → Needs Review → Approved.

Recovery

Reviewer may approve the new version as historical instead if dates or business context require.

Final state

Customer shows the newer active policy and previous version in history.

Audit history

- upload
- approval
- new version created

- active policy replaced

28.14 Rejection because document belongs to wrong customer

Starting state

Document was uploaded under the wrong customer.

User actions

1. Open review.
2. Observe critical name conflict.
3. Reject with Wrong Customer.
4. Upload the document under the correct customer.

System behavior

- Approval remains blocked.
- Rejection does not update either customer.
- Re-upload triggers duplicate warning because the file exists elsewhere.
- User confirms intentional upload after inspecting the previous association.

Visible statuses

Original: Needs Review → Rejected.

New: Pending duplicate confirmation → Processing.

Recovery

Upload under correct customer.

Final state

Wrong customer remains unchanged. Correct customer receives the new workflow.

Audit history

- customer conflict
- rejection
- duplicate warning on new upload
- intentional confirmation

28.15 Demo workflow using stored results

Starting state

Controlled demo user enters a fresh synthetic session.

User actions

1. Open sample customer.

2. Start sample document workflow.
3. Choose stored-result demonstration.
4. Review warnings.
5. Correct one field.
6. Approve.
7. Simulate email.
8. Reset demo.

System behavior

- No live AI call occurs.
- Processing stages are represented clearly as replayed.
- Review and approval behave normally.
- Email is simulated.
- Reset restores starting data.

Visible statuses

Processing marked Stored Result → Needs Review → Approved → Reset state.

Recovery

Reset at any point.

Final state

No real data or lasting public changes remain.

Audit history

Demo-only events within the session.

28.16 Usage limit reached

Starting state

Live-processing allowance is exhausted.

User actions

1. Open Upload Document.
2. Attempt live processing.
3. Open Usage.

System behavior

- Live processing action is disabled.
- Existing documents remain accessible.
- Stored demo workflows remain available.
- Usage shows Limit Reached.

Visible statuses

No new live document enters Processing.

Recovery

Wait for the configured reset, use stored results, or change operational allowance outside the normal staff workflow.

Final state

No unexpected paid attempt occurs.

Audit history

No processing event is created for a blocked action.

28.17 Email acknowledgement succeeds**Starting state**

Document is approved and customer has a valid email.

User actions

1. Select Send Acknowledgement.
2. Review preview.
3. Confirm Send.

System behavior

- Send action enters progress.
- Success is confirmed.
- Approved policy remains unchanged.

Visible statuses

Email: Not Sent → Sending → Sent.

Recovery

No recovery needed.

Final state

Send time and recipient appear.

Audit history

- email requested

- email sent

28.18 Email acknowledgement fails after approval

Starting state

Document is approved.

User actions

1. Send acknowledgement.
2. Observe failure.
3. Retry later.

System behavior

- Email failure is shown.
- Approval remains valid.
- Retry is available.
- Duplicate send warning is not shown unless a later attempt succeeds.

Visible statuses

Policy remains Approved. Email shows Failed.

Recovery

Limited resend.

Final state

Approved policy remains unchanged whether resend succeeds or fails.

Audit history

- email requested
- email failed
- resend outcome

29. Scope-creep findings

29.1 Batch upload

Why attractive

Agencies may process several declaration pages together.

Why unnecessary now

It introduces partial failures, several customer associations, combined cost confirmation, and batch progress management.

Future release

Release 2 or later after single-document reliability is proven.

29.2 Document reassignment between customers

Why attractive

It would make wrong-customer uploads easier to correct.

Why unnecessary now

It creates audit, duplicate, approval, and customer-ownership complexity.

Future release

Release 2 operational improvements.

29.3 Review assignment and locking

Why attractive

Teams may want clear reviewer ownership.

Why unnecessary now

Release 1 can safely use stale-data protection without queues, ownership, or workload routing.

Future release

Team Operations release.

29.4 Arbitrary policy-version comparison

Why attractive

Users may want to compare any two declarations.

Why unnecessary now

Release 1 only needs comparison when a proposed approval may replace the current active version.

Future release

Advanced policy-history release.

29.5 General policy editing

Why attractive

Staff may want to update policy records without another review.

Why unnecessary now

It weakens traceability and could turn the product into a general agency management system.

Future release

Only after a controlled manual-policy workflow is defined.

29.6 Deactivate or cancel policy

Why attractive

Agencies track cancelled or inactive policies.

Why unnecessary now

The product does not verify carrier status and should not imply that it does.

Future release

Carrier or agency-system integration release.

29.7 Arbitrary customer emails

Why attractive

Staff may want to customize messages.

Why unnecessary now

It creates a messaging product, template management, content risks, and send abuse.

Future release

Communications release.

29.8 Public user uploads

Why attractive

Portfolio visitors could test their own documents.

Why unnecessary now

It creates privacy, abuse, unsupported-content, storage, and cost risks.

Future release

Controlled customer trial or secure intake portal.

29.9 Field-level comments and collaboration

Why attractive

Reviewers could explain corrections to coworkers.

Why unnecessary now

There is no assignment or multi-stage review workflow.

Future release

Team collaboration release.

29.10 Automatic learning from corrections

Why attractive

The AI could improve from reviewer changes.

Why unnecessary now

It requires evaluation, training-data governance, feedback quality control, and model-change safety.

Future release

AI optimization release.

29.11 Configurable warning rules

Why attractive

Different agencies may treat fields differently.

Why unnecessary now

Release 1 supports one narrow workflow and one field set.

Future release

Multi-agency configuration release.

29.12 Advanced dashboard metrics

Why attractive

Accuracy rates, processing time, and reviewer productivity look impressive.

Why unnecessary now

A small portfolio dataset would make many metrics misleading.

Future release

Analytics release after real usage exists.

29.13 Downloadable spreadsheet or export

Why attractive

Agencies commonly work with spreadsheets.

Why unnecessary now

The approved customer and policy record already demonstrates the end-to-end workflow.

Future release

Integration and export release.

29.14 Customer self-service upload portal

Why attractive

Customers could upload declaration pages directly.

Why unnecessary now

It introduces public identity, upload security, customer communication, and consent behavior.

Future release

Connected Voice and Customer Intake release.

29.15 Voice AI

Why attractive

It strengthens the connected AI operations story.

Why unnecessary now

It introduces a separate set of conversational, audio, latency, and cost failures before the document workflow is polished.

Future release

Release 2.

30. Open functional decisions

Only the following decisions require approval before UX and architecture planning continue.

Decision 1: Customer email requirement

Question

Should customer email remain optional during customer creation?

Recommended choice

Keep email optional. Require it only before acknowledgement email can be sent.

Alternatives

- Require email for every customer.
- Remove acknowledgement email when no email exists.

Trade-offs

Optional email allows the document workflow to begin with incomplete intake information. Some customers will not be immediately eligible for acknowledgement.

Effect on scope

Minimal.

Effect on later architecture

Customer contact information must support missing values and later updates.

Decision 2: Customer contact address update during approval

Question

Should reviewers be able to update the customer's contact address from the approved declaration?

Recommended choice

Allow an explicit approval-time option. Never update automatically.

Alternatives

- Never update customer contact address.
- Always replace it with the approved mailing address.

Trade-offs

The explicit option adds one approval decision but prevents silent customer-data changes.

Effect on scope

Small addition to approval summary.

Effect on later architecture

Customer contact information and policy mailing address must remain distinguishable.

Decision 3: Exact upload limits

Question

Approve the proposed limits of 15 MB and 10 pages per PDF?

Recommended choice

Approve these Release 1 defaults.

Alternatives

- Lower limits for stricter cost control.
- Higher limits for broader document acceptance.
- Display configurable limits without fixed Release 1 defaults.

Trade-offs

The recommended limits are generous enough for declaration pages while reducing abuse and multi-document uploads.

Effect on scope

No major feature effect.

Effect on later architecture

Processing and storage must enforce and report the limits.

Decision 4: Public demo upload behavior

Question

Should public demo users be restricted to curated sample documents?

Recommended choice

Yes. Do not allow arbitrary local uploads in public demo mode.

Alternatives

- Allow one live arbitrary upload per session.
- Allow uploads but disable AI processing.
- Allow unrestricted uploads within file limits.

Trade-offs

Sample-only demo is safer and more repeatable but provides less visitor experimentation.

Effect on scope

Reduces privacy, storage, moderation, and cost behavior.

Effect on later architecture

Demo sessions can use seeded synthetic assets and controlled processing paths.

Decision 5: Demo session isolation

Question

Should each controlled demo session receive isolated disposable changes?

Recommended choice

Yes, where practical.

Alternatives

- One shared resettable demo workspace.
- Read-only demo except during owner-led demonstrations.

Trade-offs

Isolation increases implementation responsibility but prevents visitors from interfering with one another.

Effect on scope

Adds demo-session lifecycle and reset behavior.

Effect on later architecture

The architecture must define how disposable demo state is separated without introducing full customer workspaces.

Decision 6: Technical retry allowance

Question

Approve two user-triggered technical retries after the initial attempt?

Recommended choice

Yes.

Alternatives

- One retry.
- Unlimited retries.
- No user-triggered retries.

Trade-offs

Two retries provide reasonable recovery while controlling cost.

Effect on scope

Adds retry count display and disabled state.

Effect on later architecture

Processing attempts must be distinguishable and limited.

Decision 7: Correction Review for approved versions

Question

Should Release 1 allow staff to reopen an approved source for human correction without another AI call?

Recommended choice

Yes. The existing approved version remains preserved until a corrected version is approved.

Alternatives

- Require reprocessing.
- Require a replacement upload.
- Allow direct editing of approved values.

Trade-offs

Correction Review adds one controlled workflow but avoids unnecessary AI cost and preserves auditability.

Effect on scope

Adds a limited post-approval review path.

Effect on later architecture

Approved versions and editable review results must remain separate concepts.

Decision 8: Mobile approval

Question

Should full approval be available on small mobile screens?

Recommended choice

No. Support viewing and status checking, but require a larger screen for full review and approval.

Alternatives

- Fully support mobile review.
- Block the entire product on mobile.

Trade-offs

Desktop-first review reduces UX complexity while preserving basic mobile access.

Effect on scope

Reduces responsive-design requirements.

Effect on later architecture

No major effect.

31. Recommended next planning document

The next planning document should be:

Release 1 UX and Interaction Specification

This document is required before the technical architecture document because the functional behavior is now defined, but several core interactions still need to be organized into screen-level user experiences.

The UX and Interaction Specification should define:

- screen hierarchy
- page layouts
- navigation placement
- review-workspace panel behavior
- field component states
- warning presentation
- approval and rejection interaction sequences
- upload interaction sequence
- processing-progress presentation
- queue row behavior
- customer and policy-version presentation
- empty, loading, error, and stale-state placement
- demo guidance
- keyboard interaction
- responsive layout behavior
- UX copy requirements
- low-fidelity screen descriptions or wireframes

It must not define:

- technical architecture
- data storage
- APIs
- provider selection
- processing technology
- repository structure
- implementation tickets
- final colors, typography, or visual brand system
- Codex prompts

After the UX and Interaction Specification is approved, the following document should be the **Release 1 System Architecture and Technical Decision Record**, using both the functional behavior and approved interaction design as inputs.